

Dynamics of Complex Systems

Yaneer Bar-Yam



Dynamics of Complex Systems

Studies in Nonlinearity

Series Editor: Robert L. Devaney

Ralph Abraham, *Dynamics: The Geometry of Behavior*

Ralph H. Abraham and Christopher D. Shaw, *Dynamics: The Geometry of Behavior*

Robert L. Devaney, *Chaos, Fractals, and Dynamics: Computer Experiments in Mathematics*

Robert L. Devaney, *A First Course in Chaotic Dynamical Systems: Theory and Experiment*

Robert L. Devaney, *An Introduction to Chaotic Dynamical Systems, Second Edition*

Robert L. Devaney, James F. Georges, Delbert L. Johnson, *Chaotic Dynamical Systems Software*

Gerald A. Edgar (ed.), *Classics on Fractals*

James Georges, Del Johnson, and Robert L. Devaney, *Dynamical Systems Software*

Michael McGuire, *An Eye for Fractals*

Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*

Nicholas B. Tufillaro, Tyler Abbott, and Jeremiah Reilly, *An Experimental Approach to Nonlinear Dynamics and Chaos*

Yaneer Bar-Yam

Dynamics of Complex Systems

 **Routledge**
Taylor & Francis Group
LONDON AND NEW YORK

First published 1997 by Westview Press

Published 2018 by Routledge
52 Vanderbilt Avenue, New York, NY 10017
2 Park Square, Milton Park, Abingdon, Oxon OX14 4RN

Routledge is an imprint of the Taylor & Francis Group, an informa business

Copyright © 1997 by Yaneer Bar-Yam

All rights reserved. No part of this book may be reprinted or reproduced or utilised in any form or by any electronic, mechanical, or other means, now known or hereafter invented, including photocopying and recording, or in any information storage or retrieval system, without permission in writing from the publishers.

Notice:

Product or corporate names may be trademarks or registered trademarks, and are used only for identification and explanation without intent to infringe.

Library of Congress Cataloging-in-Publication Data

Bar-Yam, Yaneer.

Dynamics of complex systems / Yaneer Bar-Yam.

p. cm.

Includes index.

ISBN 0-201-55748-7

1. Biomathematics. 2. System theory. I. Title.

QH323.5.B358 1997

570'.15'1—DC21

96-52033

CIP

ISBN 13: 978-0-367-00510-8 (hbk)

This book is dedicated with love to my family

Zvi, Miriam, Aureet and Sageet

Naomi

and our children

Shlomiya, Yavni, Maayan and Taeer

Aureet's memory is a blessing.



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>



Contents

Preface xi

Acknowledgments xv

0 Overview: The Dynamics of Complex Systems—Examples, Questions, Methods and Concepts 1

- 0.1 The Field of Complex Systems 1
- 0.2 Examples 2
- 0.3 Questions 6
- 0.4 Methods 8
- 0.5 Concepts: Emergence and Complexity 9
- 0.6 For the Instructor 14

1 Introduction and Preliminaries 16

- 1.1 Iterative Maps (and Chaos) 19
- 1.2 Stochastic Iterative Maps 38
- 1.3 Thermodynamics and Statistical Mechanics 58
- 1.4 Activated Processes (and Glasses) 95
- 1.5 Cellular Automata 112
- 1.6 Statistical Fields 145
- 1.7 Computer Simulations (Monte Carlo, Simulated Annealing) 186
- 1.8 Information 214
- 1.9 Computation 235
- 1.10 Fractals, Scaling and Renormalization 258

2 Neural Networks I: Subdivision and Hierarchy 295

- 2.1 Neural Networks: Brain and Mind 296
- 2.2 Attractor Networks 300

- 2.3 Feedforward Networks 322
- 2.4 Subdivided Neural Networks 328
- 2.5 Analysis and Simulations of Subdivided Networks 345
- 2.6 From Subdivision to Hierarchy 364
- 2.7 Subdivision as a General Phenomenon 366

3 Neural Networks II: Models of Mind 371

- 3.1 Sleep and Subdivision Training 372
- 3.2 Brain Function and Models of Mind 393

4 Protein Folding I: Size Scaling of Time 420

- 4.1 The Protein-Folding Problem 421
- 4.2 Introduction to the Models 427
- 4.3 Parallel Processing in a Two-Spin Model 432
- 4.4 Homogeneous Systems 435
- 4.5 Inhomogeneous Systems 458
- 4.6 Conclusions 471

5 Protein Folding II: Kinetic Pathways 472

- 5.1 Phase Space Channels as Kinetic Pathways 473
- 5.2 Polymer Dynamics: Scaling Theory 477
- 5.3 Polymer Dynamics: Simulations 488
- 5.4 Polymer Collapse 503

6 Life I: Evolution—Origin of Complex Organisms 528

- 6.1 Living Organisms and Environments 529
- 6.2 Evolution Theory and Phenomenology 531
- 6.3 Genome, Phenome and Fitness 542
- 6.4 Exploration, Optimization and Population Interactions 550
- 6.5 Reproduction and Selection by Resources and Predators 576
- 6.6 Collective Evolution: Genes, Organisms and Populations 604
- 6.7 Conclusions 619

7 Life II: Developmental Biology—Complex by Design	621
7.1 Developmental Biology: Programming a Brick	621
7.2 Differentiation: Patterns in Animal Colors	626
7.3 Developmental Tool Kit	678
7.4 Theory, Mathematical Modeling and Biology	688
7.5 Principles of Self-Organization as Organization by Design	691
7.6 Pattern Formation and Evolution	695
8 Human Civilization I: Defining Complexity	699
8.1 Motivation	699
8.2 Complexity of Mathematical Models	705
8.3 Complexity of Physical Systems	716
8.4 Complexity Estimation	759
9 Human Civilization II: A Complex(ity) Transition	782
9.1 Introduction: Complex Systems and Social Policy	783
9.2 Inside a Complex System	788
9.3 Is Human Civilization a Complex System?	791
9.4 Toward a Networked Global Economy	796
9.5 Consequences of a Transition in Complexity	815
9.6 Civilization Itself	822
Additional Readings	827
Index	839



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

Preface

“Complex” is a word of the times, as in the often-quoted “growing complexity of life.” Science has begun to try to understand complexity in nature, a counterpoint to the traditional scientific objective of understanding the fundamental simplicity of laws of nature. It is believed, however, that even in the study of complexity there exist simple and therefore comprehensible laws. The field of study of complex systems holds that the dynamics of complex systems are founded on universal principles that may be used to describe disparate problems ranging from particle physics to the economics of societies. A corollary is that transferring ideas and results from investigators in hitherto disparate areas will cross-fertilize and lead to important new results.

In this text we introduce several of the problems of science that embody the concept of complex dynamical systems. Each is an active area of research that is at the forefront of science. Our presentation does not try to provide a comprehensive review of the research literature available in each area. Instead we use each problem as an opportunity for discussing fundamental issues that are shared among all areas and therefore can be said to unify the study of complex systems.

We do not expect it to be possible to provide a succinct definition of a complex system. Instead, we give examples of such systems and provide the elements of a definition. It is helpful to begin by describing some of the attributes that characterize complex systems. Complex systems contain a large number of mutually interacting parts. Even a few interacting objects can behave in complex ways. However, the complex systems that we are interested in have more than just a few parts. And yet there is generally a limit to the number of parts that we are interested in. If there are too many parts, even if these parts are strongly interacting, the properties of the system become the domain of conventional thermodynamics—a uniform material.

Thus far we have defined complex systems as being within the mesoscopic domain—containing more than a few, and less than too many parts. However, the mesoscopic regime describes any physical system on a particular length scale, and this is too broad a definition for our purposes. Another characteristic of most complex dynamical systems is that they are in some sense purposive. This means that the dynamics of the system has a definable objective or function. There often is some sense in which the systems are engineered. We address this topic directly when we discuss and contrast self-organization and organization by design.

A central goal of this text is to develop models and modeling techniques that are useful when applied to all complex systems. For this we will adopt both analytic tools and computer simulation. Among the analytic techniques are statistical mechanics and stochastic dynamics. Among the computer simulation techniques are cellular automata and Monte Carlo. Since analytic treatments do not yield complete theories of complex systems, computer simulations play a key role in our understanding of how these systems work.

The human brain is an important example of a complex system formed out of its component neurons. Computers might similarly be understood as complex interacting systems of transistors. Our brains are well suited for understanding complex sys-

tems, but not for simulating them. Why are computers better suited to simulations of complex systems? One could point to the need for precision that is the traditional domain of the computer. However, a better reason would be the difficulty the brain has in keeping track of many and arbitrary interacting objects or events—we can typically remember seven independent pieces of information at once. The reasons for this are an important part of the design of the brain that make it powerful for other purposes. The architecture of the brain will be discussed beginning in Chapter 2.

The study of the dynamics of complex systems creates a host of new interdisciplinary fields. It not only breaks down barriers between physics, chemistry and biology, but also between these disciplines and the so-called soft sciences of psychology, sociology, economics, and anthropology. As this breakdown occurs it becomes necessary to introduce or adopt a new vocabulary. Included in this new vocabulary are words that have been considered taboo in one area while being extensively used in another. These must be adopted and adapted to make them part of the interdisciplinary discourse. One example is the word “mind.” While the field of biology studies the brain, the field of psychology considers the mind. However, as the study of neural networks progresses, it is anticipated that the function of the neural network will become identified with the concept of mind.

Another area in which science has traditionally been mute is in the concept of meaning or purpose. The field of science traditionally has no concept of values or valuation. Its objective is to describe natural phenomena without assigning positive or negative connotation to the description. However, the description of complex systems requires a notion of purpose, since the systems are generally purposive. Within the context of purpose there may be a concept of value and valuation. If, as we will attempt to do, we address society or civilization as a complex system and identify its purpose, then value and valuation may also become a concept that attains scientific significance. There are even further possibilities of identifying value, since the very concept of complexity allows us to identify value with complexity through its difficulty of replacement. As is usual with any scientific advance, there are both dangers and opportunities with such developments.

Finally, it is curious that the origin and fate of the universe has become an accepted subject of scientific discourse—cosmology and the big bang theory—while the fate of humankind is generally the subject of religion and science fiction. There are exceptions to this rule, particularly surrounding the field of ecology—limits to population growth, global warming—however, this is only a limited selection of topics that could be addressed. Overcoming this limitation may be only a matter of having the appropriate tools. Developing the tools to address questions about the dynamics of human civilization is appropriate within the study of complex systems. It should also be recognized that as science expands to address these issues, science itself will change as it redefines and changes other fields.

Different fields are often distinguished more by the type of questions they ask than the systems they study. A significant effort has been made in this text to articulate questions, though not always to provide complete answers, since questions that define the field of complex systems will inspire more progress than answers at this early stage in the development of the field.

Like other fields, the field of complex systems has many aspects, and any text must make choices about which material to include. We have suggested that complex systems have more than a few parts and less than too many of them. There are two approaches to this intermediate regime. The first is to consider systems with more than a few parts, but still a denumerable number—denumerable, that is, by a single person in a reasonable amount of time. The second is to consider many parts, but just fewer than too many. In the first approach the main task is to describe the behavior of a particular system and its mechanism of operation—the function of a neural network of a few to a few hundred neurons, a few-celled organism, a small protein, a few people, etc. This is done by describing completely the role of each of the parts. In the second approach, the precise number of parts is not essential, and the main task is a statistical study of a collection of systems that differ from each other but share the same structure—an ensemble of systems. This approach treats general properties of proteins, neural networks, societies, etc. In this text, we adopt the second approach. However, an interesting twist to our discussion is that we will show that any complex system requires a description as a particular few-part system. A complementary volume to the present one would consider examples of systems with only a few parts and analyze their function with a view toward extracting general principles. These principles would complement the seemingly more general analysis of the statistical approach.

The order of presentation of the topics in this text is a matter of taste. Many of the chapters are self-contained discussions of a particular system or question. The first chapter contains material that provides a foundation for the rest. Part of the role of this chapter is the introduction of “simple” models upon which the remainder of the text is based. Another role is the review of concepts and techniques that will be used in later chapters so that the text is more self-contained. Because of the interdisciplinary nature of the subject matter, the first chapter is considered to have particular importance. Some of the material should be familiar to most graduate students, while other material is found only in the professional literature. For example, basic probability theory is reviewed, as well as the concepts and properties of cellular automata. The purpose is to enable this text to be read by students and researchers with a variety of backgrounds. However, it should be apparent that digesting the variety of concepts after only a brief presentation is a difficult task. Additional sources of material are listed at the end of this text.

Throughout the book, we have sought to limit advanced formal discussions to a minimum. When possible, we select models that can be described with a simpler formalism than must be used to treat the most general case possible. Where additional layers of formalism are particularly appropriate, reference is made to other literature. Simulations are described at a level of detail that, in most cases, should enable the student to perform and expand upon the simulations described. The graphical display of such simulations should be used as an integral part of exposure to the dynamics of these systems. Such displays are generally effective in developing an intuition about what are the important or relevant properties of these systems.



Taylor & Francis

Taylor & Francis Group

<http://taylorandfrancis.com>

Acknowledgments

This book is a composite of many ideas and reflects the efforts of many individuals that would be impossible to acknowledge. My personal efforts to compose this body of knowledge into a coherent framework for future study are also indebted to many who contributed to my own development. It is the earliest teachers, who we can no longer identify by memory, who should be acknowledged at the completion of a major effort. They and the teachers I remember from elementary school through graduate school, especially my thesis advisor John Joannopoulos, have my deepest gratitude. Consistent with their dedication, may this be a reward for their efforts.

The study of complex systems is a new endeavor, and I am grateful to a few colleagues and teachers who have inspired me to pursue this path. Charles Bennett through a few joint car trips opened my mind to the possibilities of this field and the paths less trodden that lead to it. Tom Malone, through his course on networked corporations, not only contributed significant concepts to the last chapter of this book, but also motivated the creation of my course on the dynamics of complex systems.

There are colleagues and students who have inspired or contributed to my understanding of various aspects of material covered in this text. Some of this contribution arises from reading and commenting on various aspects of this text, or through discussions of the material that eventually made its way here. In some cases the discussions were originally on unrelated matters, but because they were eventually connected to these subjects, they are here acknowledged. Roughly divided by area in correspondence with the order they appear in the text these include: Glasses—David Adler; Cellular Automata—Gerard Vichniac, Tom Toffoli, Norman Margolus, Mike Biafore, Eytan Domany, Danny Kandel; Computation—Jeff Siskind; Multigrid—Achi Brandt, Shlomi Taasan, Sorin Costiner; Neural Networks—John Hopfield, Sageet Bar-Yam, Tom Kincaid, Paul Appelbaum, Charles Yang, Reza Sadr-Lahijany, Jason Redi, Lee-Peng Lee, Hua Yang, Jerome Kagan, Ernest Hartmann; Protein Folding—Elisha Haas, Charles DeLisi, Temple Smith, Robert Davenport, David Mukamel, Mehran Kardar; Polymer Dynamics—Yitzhak Rabin, Mark Smith, Boris Ostrovsky, Gavin Crooks, Eliana DeBernardez-Clark; Evolution—Alan Perelson, Derren Pierre, Daniel Goldman, Stuart Kauffman, Les Kaufman; Developmental Biology—Irving Epstein, Lee Segel, Ainat Rogel, Evelyn Fox Keller; Complexity—Charles Bennett, Michael Werman, Michel Baranger; Human Economies and Societies—Tom Malone, Harry Bloom, Benjamin Samuels, Kosta Tsipis, Jonathan King.

A special acknowledgment is necessary to the students of my course from Boston University and MIT. Among them are students whose projects became incorporated in parts of this text and are mentioned above. The interest that my colleagues have shown by attending and participating in the course has brightened it for me and their contributions are meaningful: Lewis Lipsitz, Michel Baranger, Paul Barbone, George Wyner, Alice Davidson, Ed Siegel, Michael Werman, Larry Rudolf and Mehran Kardar.

Among the readers of this text I am particularly indebted to the detailed comments of Bruce Boghosian, and the supportive comments of the series editor Bob Devaney. I am also indebted to the support of Charles Cantor and Jerome Kagan.

I would like to acknowledge the constructive efforts of the editors at Addison-Wesley starting from the initial contact with Jack Repcheck and continuing with Jeff Robbins. I thank Lynne Reed for coordinating production, and at Carlisle Communications: Susan Steines, Bev Kraus, Faye Schilling, and Kathy Davis.

The software used for the text, graphs, figures and simulations of this book, includes: Microsoft Excel and Word, Deneba Canvas, Wolfram's Mathematica, and Symantec C. The hardware includes: Macintosh Quadra, and IBM RISC workstations.

The contributions of my family, to whom this book is dedicated, cannot be described in a few words.

תושלב"ע

Yaneer Bar-Yam
Newton, Massachusetts, June 1997

0

Overview:

The Dynamics of Complex Systems — Examples, Questions, Methods and Concepts

0.1 The Field of Complex Systems

The study of complex systems in a unified framework has become recognized in recent years as a new scientific discipline, the ultimate of interdisciplinary fields. It is strongly rooted in the advances that have been made in diverse fields ranging from physics to anthropology, from which it draws inspiration and to which it is relevant.

Many of the systems that surround us are complex. The goal of understanding their properties motivates much if not all of scientific inquiry. Despite the great complexity and variety of systems, universal laws and phenomena are essential to our inquiry and to our understanding. The idea that all matter is formed out of the same building blocks is one of the original concepts of science. The modern manifestation of this concept—atoms and their constituent particles—is essential to our recognition of the commonality among systems in science. The universality of constituents complements the universality of mechanical laws (classical or quantum) that govern their motion. In biology, the common molecular and cellular mechanisms of a large variety of organisms form the basis of our studies. However, even more universal than the constituents are the dynamic processes of variation and selection that in some manner cause organisms to evolve. Thus, **all scientific endeavor is based, to a greater or lesser degree, on the existence of universality**, which manifests itself in diverse ways. In this context, **the study of complex systems as a new endeavor strives to increase our ability to understand the universality that arises when systems are highly complex.**

A dictionary definition of the word “complex” is: “consisting of interconnected or interwoven parts.” Why is the nature of a complex system inherently related to its parts? Simple systems are also formed out of parts. To explain the difference between simple and complex systems, the terms “interconnected” or “interwoven” are somehow essential. Qualitatively, to understand the behavior of a complex system we must understand not only the behavior of the parts but how they act together to form the behavior of the whole. It is because we cannot describe the whole without describing each part, and because each part must be described in relation to other parts, that complex systems are difficult to understand. This is relevant to another definition of “complex”: “not easy to understand or analyze.” These qualitative ideas about what a complex system is can be made more quantitative. Articulating them in a clear way is

both essential and fruitful in pointing the way toward progress in understanding the universal properties of these systems.

For many years, professional specialization has led science to progressive isolation of individual disciplines. How is it possible that well-separated fields such as molecular biology and economics can suddenly become unified in a single discipline? How does the study of complex systems in general pertain to the detailed efforts devoted to the study of particular complex systems? In this regard one must be careful to acknowledge that there is always a dichotomy between universality and specificity. A study of universal principles does not replace detailed description of particular complex systems. However, universal principles and tools guide and simplify our inquiries into the study of specifics. For the study of complex systems, universal simplifications are particularly important. **Sometimes universal principles are intuitively appreciated without being explicitly stated.** However, a careful articulation of such principles can enable us to approach particular systems with a systematic guidance that is often absent in the study of complex systems.

A pictorial way of illustrating the relationship of the field of complex systems to the many other fields of science is indicated in Fig. 0.1.1. This figure shows the conventional view of science as progressively separating into disparate disciplines in order to gain knowledge about the ever larger complexity of systems. It also illustrates the view of the field of complex systems, which suggests that all complex systems have universal properties. Because each field develops tools for addressing the complexity of the systems in their domain, many of these tools can be adapted for more general use by recognizing their universal applicability. Hence the motivation for cross-disciplinary fertilization in the study of complex systems.

In Sections 0.2–0.4 we initiate our study of complex systems by discussing examples, questions and methods that are relevant to the study of complex systems. Our purpose is to introduce the field without a strong bias as to conclusions, so that the student can develop independent perspectives that may be useful in this new field—opening the way to his or her own contributions to the study of complex systems. In Section 0.5 we introduce two key concepts—emergence and complexity—that will arise through our study of complex systems in this text.

0.2 Examples

0.2.1 A few examples

What are complex systems and what properties characterize them? It is helpful to start by making a list of some examples of complex systems. Take a few minutes to make your own list. Consider actual systems rather than mathematical models (we will consider mathematical models later). Make a list of some simple things to contrast them with.

Examples of Complex Systems

Governments

Families

The human body—physiological perspective

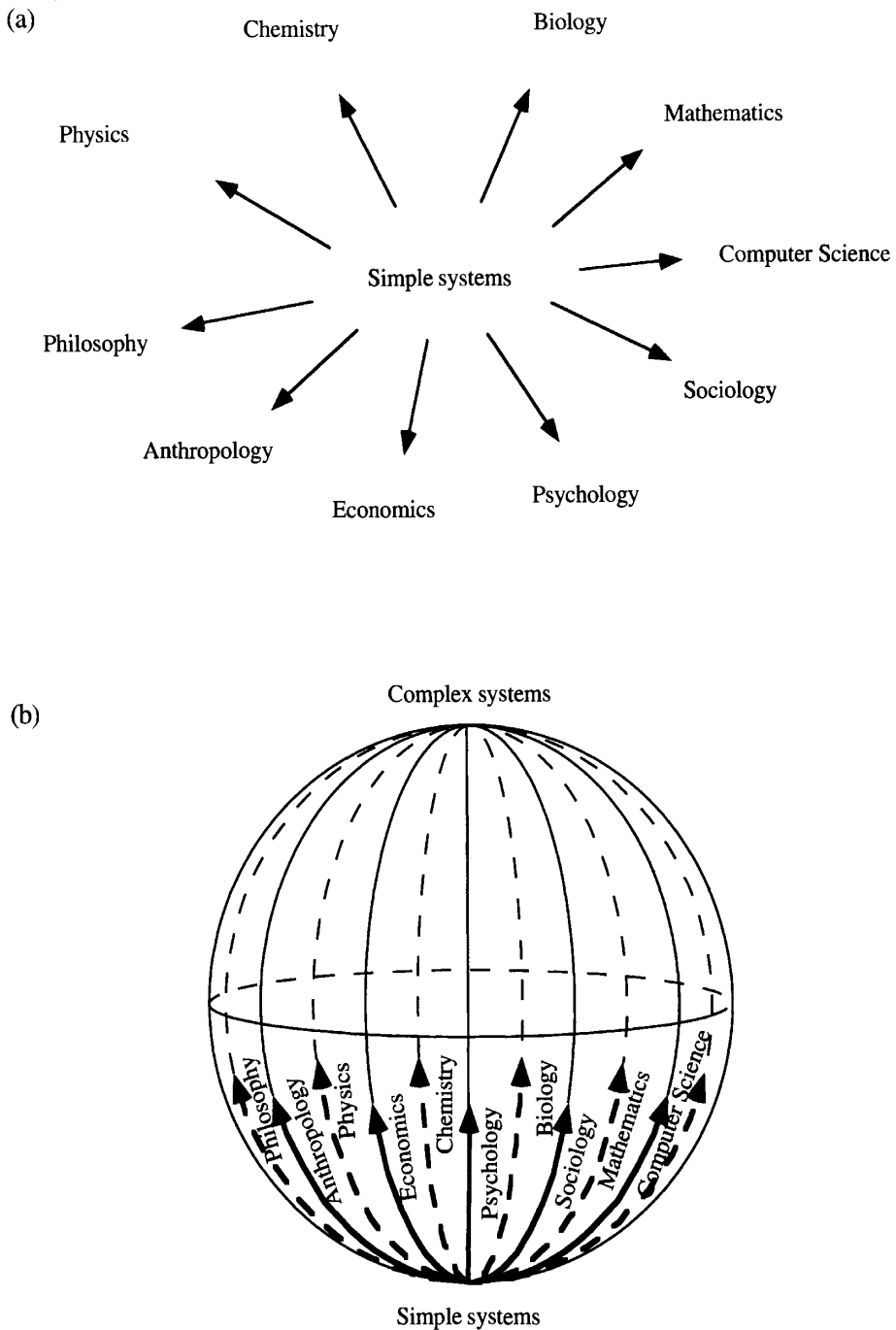


Figure 0.1.1 Conceptual illustration of the space of scientific inquiry. (a) is the conventional view where disciplines diverge as knowledge increases because of the increasing complexity of the various systems being studied. In this view all knowledge is specific and knowledge is gained by providing more and more details. (b) illustrates the view of the field of complex systems where complex systems have universal properties. By considering the common properties of complex systems, one can approach the specifics of particular complex systems from the top of the sphere as well as from the bottom.

A person—psychosocial perspective
 The brain
 The ecosystem of the world
 Subworld ecosystems: desert, rain forest, ocean
 Weather
 A corporation
 A computer

Examples of Simple Systems

An oscillator
 A pendulum
 A spinning wheel
 An orbiting planet

The purpose of thinking about examples is to develop a first understanding of the question, What makes systems complex? To begin to address this question we can start describing systems we know intuitively as complex and see what properties they share. We try this with the first two examples listed above as complex systems.

Government

- It has many different functions: military, immigration, taxation, income distribution, transportation, regulation. Each function is itself complex.
- There are different levels and types of government: local, state and federal; town meeting, council, mayoral. There are also various governmental forms in different countries.

Family

- It is a set of individuals.
- Each individual has a relationship with the other individuals.
- There is an interplay between the relationship and the qualities of the individual.
- The family has to interact with the outside world.
- There are different kinds of families: nuclear family, extended family, etc.

These descriptions focus on function and structure and diverse manifestation. We can also consider the role that time plays in complex systems. Among the properties of complex systems are change, growth and death, possibly some form of life cycle. Combining time and the environment, we would point to the ability of complex systems to adapt.

One of the issues that we will need to address is whether there are different categories of complex systems. For example, we might contrast the systems we just described with complex physical systems: hydrodynamics (fluid flow, weather), glasses, composite materials, earthquakes. In what way are these systems similar to or different from the biological or social complex systems? Can we assign function and discuss structure in the same way?

0.2.2 *Central properties of complex systems*

After beginning to describe complex systems, a second step is to identify commonalities. We might make a list of some of the characteristics of complex systems and assign each of them some measure or attribute that can provide a first method of classification or description.

- Elements (and their number)
- Interactions (and their strength)
- Formation/Operation (and their time scales)
- Diversity/Variability
- Environment (and its demands)
- Activity(ies) (and its[their] objective[s])

This is a first step toward quantifying the properties of complex systems. Quantifying the last three in the list requires some method of counting possibilities. The problem of counting possibilities is central to the discussion of quantitative complexity.

0.2.3 *Emergence: From elements and parts to complex systems*

There are two approaches to organizing the properties of complex systems that will serve as the foundation of our discussions. The first of these is the relationship between elements, parts and the whole. Since there is only one property of the complex system that we know for sure — that it is complex—the primary question we can ask about this relationship is **how the complexity of the whole is related to the complexity of the parts**. As we will see, this question is a compelling question for our understanding of complex systems.

From the examples we have indicated above, it is apparent that parts of a complex system are often complex systems themselves. This is reasonable, because when the parts of a system are complex, it seems intuitive that a collection of them would also be complex. However, this is not the only possibility.

Can we describe a system composed of simple parts where the collective behavior is complex? This is an important possibility, called emergent complexity. Any complex system formed out of atoms is an example. The idea of emergent complexity is that the behaviors of many simple parts interact in such a way that the behavior of the whole is complex. Elements are those parts of a complex system that may be considered simple when describing the behavior of the whole.

Can we describe a system composed of complex parts where the collective behavior is simple? This is also possible, and it is called emergent simplicity. A useful example is a planet orbiting around a star. The behavior of the planet is quite simple, even if the planet is the Earth, with many complex systems upon it. This example illustrates the possibility that the collective system has a behavior at a different scale than its parts. On the smaller scale the system may behave in a complex way, but on the larger scale all the complex details may not be relevant.

0.2.4 What is complexity?

The second approach to the study of complex systems begins from an understanding of the relationship of systems to their descriptions. The central issue is defining quantitatively what we mean by complexity. What, after all, do we mean when we say that a system is complex? Better yet, what do we mean when we say that one system is more complex than another? Is there a way to identify the complexity of one system and to compare it with the complexity of another system? To develop a quantitative understanding of complexity we will use tools of both statistical physics and computer science—information theory and computation theory. According to this understanding, **complexity is the amount of information necessary to describe a system.** However, in order to arrive at a consistent definition, care must be taken to specify the level of detail provided in the description.

One of our targets is to understand how this concept of complexity is related to emergence—emergent complexity and emergent simplicity. Can we understand why information-based complexity is related to the description of elements, and how their behavior gives rise to the collective complexity of the whole system?

Section 0.5 of this overview discusses further the concepts of emergence and complexity, providing a simplified preview of the more complete discussions later in this text.

0.3 Questions

This text is structured around four questions related to the characterization of complex systems:

1. Space: What are the characteristics of the structure of complex systems? Many complex systems have substructure that extends all the way to the size of the system itself. Why is there substructure?
2. Time: How long do dynamical processes take in complex systems? Many complex systems have specific responses to changes in their environment that require changing their internal structure. How can a complex structure respond in a reasonable amount of time?
3. Self-organization and/versus organization by design: How do complex systems come into existence? What are the dynamical processes that can give rise to complex systems? Many complex systems undergo guided developmental processes as part of their formation. How are developmental processes guided?
4. Complexity: What is complexity? Complex systems have varying degrees of complexity. How do we characterize/distinguish the varying degrees of complexity?

Chapter 1 of this text plays a special role. Its ten sections introduce mathematical tools. These tools and their related concepts are integral to our understanding of complex system behavior. The main part of this book consists of eight chapters, 2–9. These

chapters are paired. Each pair discusses one of the above four questions in the context of a particular complex system. Chapters 2 and 3 discuss the role of substructure in the context of neural networks. Chapters 4 and 5 discuss the time scale of dynamics in the context of protein folding. Chapters 6 and 7 discuss the mechanisms of organization of complex systems in the context of living organisms. Chapters 8 and 9 discuss complexity in the context of human civilization. In each case the first of the pair of chapters discusses more general issues and models. The second tends to be more specialized to the system that is under discussion. There is also a pattern to the degree of analytic, simulation or qualitative treatments. In general, the first of the two chapters is more analytic, while the second relies more on simulations or qualitative treatments. Each chapter has at least some discussion of qualitative concepts in addition to the formal quantitative discussion.

Another way to regard the text is to distinguish between the two approaches summarized above. The first deals with elements and interactions. The second deals with descriptions and information. Ultimately, our objective is to relate them, but we do so using questions that progress gradually from the elements and interactions to the descriptions and information. The former dominates in earlier chapters, while the latter is important for Chapter 6 and becomes dominant in Chapters 8 and 9.

While the discussion in each chapter is presented in the context of a specific complex system, our focus is on complex systems in general. Thus, we do not attempt (nor would it be possible) to review the entire fields of neural networks, protein folding, evolution, developmental biology and social and economic sciences. Since we are interested in universal aspects of these systems, the topics we cover need not be the issues of contemporary importance in the study of these systems. Our approach is to motivate a question of interest in the context of complex systems using a particular complex system, then to step back and adopt a method of study that has relevance to all complex systems. Researchers interested in a particular complex system are as likely to find a discussion of interest to them in any one of the chapters, and should not focus on the chapter with the particular complex system in its title.

We note that the text is interrupted by questions that are, with few exceptions, solved in the text. They are given as questions to promote independent thought about the study of complex systems. Some of them develop further the analysis of a system through analytic work or through simulations. Others are designed for conceptual development. With few exceptions they should be considered integral to the text, and even if they are not solved by the reader, the solutions should be read.

Question 0.3.1 Consider a few complex systems. Make a list of their elements, interactions between these elements, the mechanism by which the system is formed and the activities in which the system is engaged.

Solution 0.3.1 The following table indicates properties of the systems that we will be discussing most intensively in this text. ■

System	Element	Interaction	Formation	Activity
Proteins	Amino Acids	Bonds	Protein folding	Enzymatic activity
Nervous system Neural networks	Neurons	Synapses	Learning	Behavior Thought
Physiology	Cells	Chemical messengers Physical support	Developmental biology	Movement Physiological functions
Life	Organisms	Reproduction Competition Predation Communication	Evolution	Survival Reproduction Consumption Excretion
Human economies and societies	Human Beings Technology	Communication Confrontation Cooperation	Social evolution	Same as Life? Exploration?

Table 0.3.1: Complex Systems and Some Attributes

0.4 Methods

When we think about methodology, we must keep purpose in mind. **Our purpose in studying complex systems is to extract general principles.** General principles can take many forms. Most principles are articulated as relationships between properties—when a system has the property x , then it has the property y . When possible, relationships should be quantitative and expressed as equations. In order to explore such relationships, we must construct and study mathematical models. Asking why the property x is related to the property y requires an understanding of alternatives. What else is possible? As a bonus, when we are able to generate systems with various properties, we may also be able to use them for practical applications.

All approaches that are used for the study of simple systems can be applied to the study of complex systems. However, it is important to recognize features of conventional approaches that may hamper progress in the study of complex systems. Both experimental and theoretical methods have been developed to overcome these difficulties. In this text we introduce and use methods of analysis and simulation that are particularly suited to the study of complex systems. These methods avoid standard simplifying assumptions, but use other simplifications that are better suited to our objectives. We discuss some of these in the following paragraphs.

- Don't take it apart. Since interactions between parts of a complex system are essential to understanding its behavior, looking at parts by themselves is not sufficient. It is necessary to look at parts in the context of the whole. Similarly, a complex system interacts with its environment, and this environmental influence is

important in describing the behavior of the system. Experimental tools have been developed for studying systems *in situ* or *in vivo*—in context. Theoretical analytic methods such as the mean field approach enable parts of a system to be studied in context. Computer simulations that treat a system in its entirety also avoid such problems.

- Don't assume smoothness. Much of the quantitative study of simple systems makes use of differential equations. Differential equations, like the wave equation, assume that a system is essentially uniform and that local details don't matter for the behavior of a system on larger scales. These assumptions are not generally valid for complex systems. Alternate static models such as fractals, and dynamical models including iterative maps and cellular automata may be used instead.
- Don't assume that only a few parameters are important. The behavior of complex systems depends on many independent pieces of information. Developing an understanding of them requires us to build mental models. However, we can only have “in mind” 7 ± 2 independent things at once. Analytic approaches, such as scaling and renormalization, have been developed to identify the few relevant parameters when this is possible. Information-based approaches consider the collection of all parameters as the object of study. Computer simulations keep track of many parameters and may be used in the study of dynamical processes.

There are also tools needed for communication of the results of studies. Conventional manuscripts and oral presentations are now being augmented by video and interactive media. Such novel approaches can increase the effectiveness of communication, particularly of the results of computer simulations. However, we should avoid the “cute picture” syndrome, where pictures are presented without accompanying discussion or analysis.

In this text, we introduce and use a variety of analytic and computer simulation methods to address the questions listed in the previous section. As mentioned in the preface, there are two general methods for studying complex systems. In the first, a specific system is selected and each of the parts as well as their interactions are identified and described. Subsequently, the objective is to show how the behavior of the whole emerges from them. The second approach considers a class of systems (ensemble), where the essential characteristics of the class are described, and statistical analysis is used to obtain properties and behaviors of the systems. In this text we focus on the latter approach.

0.5 Concepts: Emergence and Complexity

The objectives of the field of complex systems are built on fundamental concepts—emergence, complexity—about which there are common misconceptions that are addressed in this section and throughout the book. Once understood, these concepts reveal the context in which universal properties of complex systems arise and specific universal phenomena, such as the evolution of biological systems, can be better understood.

A complex system is a system formed out of many components whose behavior is emergent, that is, the behavior of the system cannot be simply inferred from the behavior of its components. The amount of information necessary to describe the behavior of such a system is a measure of its complexity. In the following sections we discuss these concepts in greater detail.

0.5.1 Emergence

It is impossible to understand complex systems without recognizing that simple atoms must somehow, in large numbers, give rise to complex collective behaviors. How and when this occurs is the simplest and yet the most profound problem that the study of complex systems faces. The problem can be approached first by developing an understanding of the term “emergence.” For many, the concept of emergent behavior means that the behavior is not captured by the behavior of the parts. This is a serious misunderstanding. It arises because the collective behavior is not readily understood from the behavior of the parts. The collective behavior is, however, contained in the behavior of the parts if they are studied in the context in which they are found. To explain this, we discuss examples of emergent properties that illustrate the difference between local emergence—where collective behavior appears in a small part of the system—and global emergence—where collective behavior pertains to the system as a whole. It is the latter which is particularly relevant to the study of complex systems.

We can speak about emergence when we consider a collection of elements and the properties of the collective behavior of these elements. In conventional physics, the main arena for the study of such properties is thermodynamics and statistical mechanics. The easiest thermodynamic system to think about is a gas of particles. Two emergent properties of a gas are its pressure and temperature. The reason they are emergent is that they do not naturally arise out of the description of an individual particle. We generally describe a particle by specifying its position and velocity. Pressure and temperature become relevant only when we have many particles together. While these are emergent properties, the way they are emergent is very limited. We call them local emergent properties. The pressure and temperature is a local property of the gas. We can take a very small sample of the gas away from the rest and still define and measure the (same) pressure and temperature. Such properties, called intensive in physics, are local emergent properties. Other examples from physics of locally emergent behavior are collective modes of excitation such as sound waves, or light propagation in a medium. Phase transitions (e.g., solid to liquid) also represent a collective dynamics that is visible on a macroscopic scale, but can be seen in a microscopic sample as well.

Another example of a local emergent property is the formation of water from atoms of hydrogen and oxygen. The properties of water are not apparent in the properties of gasses of oxygen or hydrogen. Neither does an isolated water molecule reveal most properties of water. However, a microscopic amount of water is sufficient.

In the study of complex systems we are particularly interested in global emergent properties. Such properties depend on the entire system. The mathematical treatment of global emergent properties requires some effort. This is one reason that emergence is not well appreciated or understood. We will discuss global emergence by summariz-

ing the results of a classic mathematical treatment, and then discuss it in a more general manner that can be readily appreciated and is useful for semiquantitative analyses.

The classic analysis of global emergent behavior is that of an associative memory in a simple model of neural networks known as the Hopfield or attractor network. The analogy to a neural network is useful in order to be concrete and relate this model to known concepts. However, this is more generally a model of any system formed from simple elements whose states are correlated. Without such correlations, emergent behavior is impossible. Yet if all elements are correlated in a simple way, then local emergent behavior is the outcome. Thus a model must be sufficiently rich in order to capture the phenomenon of global emergent behavior. One of the important qualities of the attractor network is that it displays global emergence in a particularly elegant manner. The following few paragraphs summarize the operation of the attractor network as an associative memory.

The Hopfield network has simple binary elements that are either ON or OFF. The binary elements are an abstraction of the firing or quiescent state of neurons. The elements interact with each other to create correlations in the firing patterns. The interactions represent the role of synapses in a neural network. The network can work as a memory. Given a set of preselected patterns, it is possible to set the interactions so that these patterns are self-consistent states of the network—the network is stable when it is in these firing patterns. Even if we change some of the neurons, the original pattern will be recovered. This is an associative memory.

Assume for the moment that the pattern of firing represents a sentence, such as “To be or not to be, that is the question.” We can recover the complete sentence by presenting only part of it to the network “To be or not to be, that” might be enough. We could use any part to retrieve the whole, such as, “to be, that is the question.” This kind of memory is to be contrasted with a computer memory, which works by assigning an address to each storage location. To access the information stored in a particular location we need to know the address. In the neural network memory, we specify part of what is located there, rather than the analogous address: Hamlet, by William Shakespeare, act 3, scene 1, line 64.

More central to our discussion, however, is that in a computer memory a particular bit of information is stored in a particular switch. By contrast, the network does not have its memory in a neuron. Instead the memory is in the synapses. In the model, there are synapses between each neuron and every other neuron. If we remove a small part of the network and look at its properties, then the number of synapses that a neuron is left with in this small part is only a small fraction of the number of synapses it started with. If there are more than a few patterns stored, then **when we cut out the small part of the network it loses the ability to remember any of the patterns, even the part which would be represented by the neurons contained in this part.**

This kind of behavior characterizes emergent properties. We see that emergent properties cannot be studied by physically taking a system apart and looking at the parts (reductionism). They can, however, be studied by looking at each of the parts in the context of the system as a whole. This is the nature of emergence and an indication of how it can be studied and understood.

The above discussion reflects the analysis of a relatively simple mathematical model of emergent behavior. We can, however, provide a more qualitative discussion that serves as a guide for thinking about diverse complex systems. This discussion focuses on the properties of a system when part of it is removed. Our discussion of local emergent properties suggested that taking a small part out of a large system would cause little change in the properties of the small part, or the properties of the large part. On the other hand, when a system has a global emergent property, the behavior of the small part is different in isolation than when it is part of the larger system.

If we think about the system as a whole, rather than the small part of the system, we can identify the system that has a global emergent property as being formed out of interdependent parts. The term “interdependent” is used here instead of the terms “interconnected” or “interwoven” used in the dictionary definition of “complex” quoted in Section 0.1, because neither of the latter terms pertain directly to the influence one part has on another, which is essential to the properties of a dynamic system. “Interdependent” is also distinct from “interacting,” because even strong interactions do not necessarily imply interdependence of behavior. This is clear from the macroscopic properties of simple solids.

Thus, we can characterize complex systems through the effect of removal of part of the system. There are two natural possibilities. The first is that properties of the part are affected, but the rest is not affected. The second is that properties of the rest are affected by the removal of a part. It is the latter that is most appealing as a model of a truly complex system. Such a system has a collective behavior that is dependent on the behavior of all of its parts. This concept becomes more precise when we connect it to a quantitative measure of complexity.

0.5.2 Complexity

The second concept that is central to complex systems is a quantitative measure of how complex a system is. Loosely speaking, the complexity of a system is the amount of information needed in order to describe it. The complexity depends on the level of detail required in the description. A more formal definition can be understood in a simple way. If we have a system that could have many possible states, but we would like to specify which state it is actually in, then the number of binary digits (bits) we need to specify this particular state is related to the number of states that are possible. If we call the number of states Ω then the number of bits of information needed is

$$I = \log_2(\Omega) \quad (0.5.1)$$

To understand this we must realize that to specify which state the system is in, we must enumerate the states. Representing each state uniquely requires as many numbers as there are states. Thus the number of states of the representation must be the same as the number of states of the system. For a string of N bits there are 2^N possible states and thus we must have

$$\Omega = 2^N \quad (0.5.2)$$

which implies that N is the same as I above. Even if we use a descriptive English text instead of numbers, there must be the same number of possible descriptions as there are states, and the information content must be the same. When the number of possible valid English sentences is properly accounted for, it turns out that the best estimate of the amount of information in English is about 1 bit per character. This means that the information content of this sentence is about 120 bits, and that of this book is about 3×10^6 bits.

For a microstate of a physical system, where we specify the positions and momenta of each of the particles, this can be recognized as proportional to the entropy of the system, which is defined as

$$S = k \ln(\Omega) = k \ln(2)I \quad (0.5.3)$$

where $k = 1.38 \times 10^{-23}$ Joule/°Kelvin is the Boltzmann constant which is relevant to our conventional choice of units. Using measured entropies we find that entropies of order 10 bits per atom are typical. The reason k is so small is that the quantities of matter we typically consider are in units of Avogadro's number (moles) and the number of bits per mole is 6.02×10^{23} times as large. Thus, the information in a piece of material is of order 10^{24} bits.

There is one point about Eq. (0.5.3) that may require some clarification. The positions and momenta of particles are real numbers whose specification might require infinitely many bits. Why isn't the information necessary to specify the microstate of a system infinite? The answer to this question comes from quantum physics, which is responsible for giving a unique value to the entropy and thus the information needed to specify a state of the system. It does this in two ways. First, it tells us that microscopic states are indistinguishable unless they differ by a discrete amount in position and momentum—a quantum difference given by Planck's constant h . Second, it indicates that particles like nuclei or atoms in their ground state are uniquely specified by this state, and are indistinguishable from each other. There is no additional information necessary to specify their internal structure. Under standard conditions, essentially all nuclei are in their lowest energy state.

The relationship of entropy and information is not accidental, of course, but it is the source of much confusion. The confusion arises because the entropy of a physical system is largest when it is in equilibrium. This suggests that the most complex system is a system in equilibrium. This is counter to our usual understanding of complex systems. Equilibrium systems have no spatial structure and do not change over time. Complex systems have substantial internal structure and this structure changes over time.

The problem is that we have used the definition of the information necessary to specify the microscopic state (microstate) of the system rather than the macroscopic state (macrostate) of the system. We need to consider the information necessary to describe the macrostate of the system in order to define what we mean by complexity. One of the important points to realize is that in order for the macrostate of the system to require a lot of information to describe it, there must be correlations in the microstate of the system. It is only when many microscopic atoms move in a coherent fashion that we can see this motion on a macroscopic scale. However, if many

microscopic atoms move together, the system must be far from equilibrium and the microscopic information (entropy) must be lower than that of an equilibrium system.

It is helpful, even essential, to define a complexity profile which is a function of the scale of observation. To obtain the complexity profile, we observe the system at a particular length (or time) scale, ignoring all finer-scale details. Then we consider how much information is necessary to describe the observations on this scale. This solves the problem of distinguishing between a microscopic and a macroscopic description. Moreover, for different choices of scale, it explicitly captures the dependence of the complexity on the level of detail that is required in the description.

The complexity profile must be a monotonically falling function of the scale. This is because the information needed to describe a system on a larger scale must be a subset of the information needed to describe the system on a smaller scale—any finer-scale description contains the coarser-scale description. The complexity profile characterizes the properties of a complex system. If we wish to point to a particular number for the complexity of a system, it is natural to consider the complexity as the value of the complexity profile at a scale that is slightly smaller than the size of the system itself. The behavior at this scale includes the movement of the system through space, and dynamical changes of the system that are essentially the size of the system as a whole. The Earth orbiting the sun is a useful example.

We can make a direct connection between this definition of complexity and the discussion of the formation of a complex system out of parts. The complexity of the parts of the system are described by the complexity profile of the system evaluated on the scale of the parts. When the behavior of the system depends on the behavior of the parts, the complexity of the whole must involve a description of the parts, thus it is large. The smaller the parts that must be described to describe the behavior of the whole, the larger the complexity of the entire system.

0.6 For the Instructor

This text is designed for use in an introductory graduate-level course, to present various concepts and methodologies of the study of complex systems and to begin to develop a common language for researchers in this new field. It has been used for a one-semester course, but the amount of material is large, and it is better to spread the material over two semesters. A two-semester course also provides more opportunities for including various other approaches to the study of complex systems, which are as valuable as the ones that are covered here and may be more familiar to the instructor.

Consistent with the objective and purpose of the field, students attending such a course tend to have a wide variety of backgrounds and interests. While this is a positive development, it causes difficulties for the syllabus and framework of the course.

One approach to a course syllabus is to include the introductory material given in Chapter 1 as an integral part of the course. It is better to interleave the later chapters with the relevant materials from Chapter 1. Such a course might proceed: 1.1–1.6; 2; 3; 4; 1.7; 5; 6; 7; 1.8–1.10; 8; 9. Including the materials of Chapter 1 allows the dis-

cussion of important mathematical methods, and addresses the diverse backgrounds of the students. Even if the introductory chapter is covered quickly (e.g., in a one-semester course), this establishes a common base of knowledge for the remainder of the course. If a high-speed approach is taken, it must be emphasized to the students that this material serves only to expose them to concepts that they are unfamiliar with, and to review concepts for those with prior knowledge of the topics covered. Unfortunately, many students are not willing to sit through such an extensive (and intense) introduction.

A second approach begins from Chapter 2 and introduces the material from Chapter 1 only as needed. The chapters that are the most technically difficult, and rely the most on Chapter 1, are Chapters 4 and 5. Thus, for a one-semester course, the subject of protein folding (Chapters 4 and 5) could be skipped. Then much of the introductory material can be omitted, with the exception of a discussion of the last part of Section 1.3, and some introduction to the subject of entropy and information either through thermodynamics (Section 1.3) or information theory (Section 1.8), preferably both. Then Chapters 2 and 3 can be covered first, followed by Chapters 6–9, with selected material introduced from Chapter 1 as is appropriate for the background of the students.

There are two additional recommendations. First, it is better to run this course as a project-based course rather than using graded homework. The varied backgrounds of students make it difficult to select and fairly grade the problems. Projects for individuals or small groups of students can be tailored to their knowledge and interests. There are many new areas of inquiry, so that projects may approach research-level contributions and be exciting for the students. Unfortunately, this means that students may not devote sufficient effort to the study of course material, and rely largely upon exposure in lectures. There is no optimal solution to this problem. Second, if it is possible, a seminar series with lecturers who work in the field should be an integral part of the course. This provides additional exposure to the varied approaches to the study of complex systems that it is not possible for a single lecturer or text to provide.

1

Introduction and Preliminaries

Conceptual Outline

■ **1.1** ■ A deceptively simple model of the dynamics of a system is a deterministic iterative map applied to a single real variable. We characterize the dynamics by looking at its limiting behavior and the approach to this limiting behavior. Fixed points that attract or repel the dynamics, and cycles, are conventional limiting behaviors of a simple dynamic system. However, changing a parameter in a quadratic iterative map causes it to undergo a sequence of cycle doublings (bifurcations) until it reaches a regime of chaotic behavior which cannot be characterized in this way. This deterministic chaos reveals the potential importance of the influence of fine-scale details on large-scale behavior in the dynamics of systems.

■ **1.2** ■ A system that is subject to complex (external) influences has a dynamics that may be modeled statistically. The statistical treatment simplifies the complex unpredictable stochastic dynamics of a single system, to the simple predictable dynamics of an ensemble of systems subject to all possible influences. A random walk on a line is the prototype stochastic process. Over time, the random influence causes the ensemble of walkers to spread in space and form a Gaussian distribution. When there is a bias in the random walk, the walkers have a constant velocity superimposed on the spreading of the distribution.

■ **1.3** ■ While the microscopic dynamics of physical systems is rapid and complex, the macroscopic behavior of many materials is simple, even static. Before we can understand how complex systems have complex behaviors, we must understand why materials can be simple. The origin of simplicity is an averaging over the fast microscopic dynamics on the time scale of macroscopic observations (the ergodic theorem) and an averaging over microscopic spatial variations. The averaging can be performed theoretically using an ensemble representation of the physical system that assumes all microscopic states are realized. Using this as an assumption, a statistical treatment of microscopic states describes the macroscopic equilibrium behavior of systems. The final part of Section 1.3 introduces concepts that play a central role in the rest of the book. It discusses the differences between equilibrium and complex systems. Equilibrium systems are divisible and satisfy the ergodic theorem. Complex systems

are composed out of interdependent parts and violate the ergodic theorem. They have many degrees of freedom whose time dependence is very slow on a microscopic scale.

1.4 To understand the separation of time scales between fast and slow degrees of freedom, a two-well system is a useful model. The description of a particle traveling in two wells can be simplified to the dynamics of a two-state (binary variable) system. The fast dynamics of the motion within a well is averaged by assuming that the system visits all states, represented as an ensemble. After taking the average, the dynamics of hopping between the wells is represented explicitly by the dynamics of a binary variable. The hopping rate depends exponentially on the ratio of the energy barrier and the temperature. When the temperature is low enough, the hopping is frozen. Even though the two wells are not in equilibrium with each other, equilibrium continues to hold within a well. The cooling of a two-state system serves as a simple model of a glass transition, where many microscopic degrees of freedom become frozen at the glass transition temperature.

1.5 Cellular automata are a general approach to modeling the dynamics of spatially distributed systems. Expanding the notion of an iterative map of a single variable, the variables that are updated are distributed on a lattice in space. The influence between variables is assumed to rely upon local interactions, and is homogeneous. Space and time are both discretized, and the variables are often simplified to include only a few possible states at each site. Various cellular automata can be designed to model key properties of physical and biological systems.

1.6 The equilibrium state of spatially distributed systems can be modeled by fields that are treated using statistical ensembles. The simplest is the Ising model, which captures the simple cooperative behavior found in magnets and many other systems. Cooperative behavior is a mechanism by which microscopic fast degrees of freedom can become slow collective degrees of freedom that violate the ergodic theorem and are visible macroscopically. Macroscopic phase transitions are the dynamics of the cooperative degrees of freedom. Cooperative behavior of many interacting elements is an important aspect of the behavior of complex systems. This should be contrasted to the two-state model (Section 1.4), where the slow dynamics occurs microscopically.

1.7 Computer simulations of models such as molecular dynamics or cellular automata provide important tools for the study of complex systems. Monte Carlo simulations enable the study of ensemble averages without necessarily describing the dynamics of a system. However, they can also be used to study random-walk dynamics. Minimization methods that use iterative progress to find a local minimum are often an important aspect of computer simulations. Simulated annealing is a method that can help find low energy states on complex energy surfaces.

1.8 We have treated systems using models without acknowledging explicitly that our objective is to describe them. All our efforts are designed to map a system onto a description of the system. For complex systems the description must be quite long, and the study of descriptions becomes essential. With this recognition, we turn

to information theory. The information contained in a communication, typically a string of characters, may be defined quantitatively as the logarithm of the number of possible messages. When different messages have distinct probabilities P in an ensemble, then the information can be identified as $-\ln(P)$ and the average information is defined accordingly. Long messages can be modeled using the same concepts as a random walk, and we can use such models to estimate the information contained in human languages such as English.

1.9 In order to understand the relationship of information to systems, we must also understand what we can infer from information that is provided. The theory of logic is concerned with inference. It is directly linked to computation theory, which is concerned with the possible (deterministic) operations that can be performed on a string of characters. All operations on character strings can be constructed out of elementary logical (Boolean) operations on binary variables. Using Turing's model of computation, it is further shown that all computations can be performed by a universal Turing machine, as long as its input character string is suitably constructed. Computation theory is also related to our concern with the dynamics of physical systems because it explores the set of possible outcomes of discrete deterministic dynamic systems.

1.10 We return to issues of structure on microscopic and macroscopic scales by studying fractals that are self-similar geometric objects that embody the concept of progressively increasing structure on finer and finer length scales. A general approach to the scale dependence of system properties is described by scaling theory. The renormalization group methodology enables the study of scaling properties by relating a model of a system on one scale with a model of the system on another scale. Its use is illustrated by application to the Ising model (Section 1.6), and to the bifurcation route to chaos (Section 1.1). Renormalization helps us understand the basic concept of modeling systems, and formalizes the distinction between relevant and irrelevant microscopic parameters. Relevant parameters are the microscopic parameters that can affect the macroscopic behavior. The concept of universality is the notion that a whole class of microscopic models will give rise to the same macroscopic behavior, because many parameters are irrelevant. A conceptually related computational technique, the multigrid method, is based upon representing a problem on multiple scales.

The study of complex systems begins from a set of models that capture aspects of the dynamics of simple or complex systems. These models should be sufficiently general to encompass a wide range of possibilities but have sufficient structure to capture interesting features. An exciting bonus is that even the apparently simple models discussed in this chapter introduce features that are not typically treated in the conventional science of simple systems, but are appropriate introductions to the dynamics of complex systems. Our treatment of dynamics will often consider discrete rather than continuous time. Analytic treatments are often convenient to formulate in continu-

ous variables and differential equations; however, computer simulations are often best formulated in discrete space-time variables with well-defined intervals. Moreover, the assumption of a smooth continuum at small scales is not usually a convenient starting point for the study of complex systems. We are also generally interested not only in one example of a system but rather in a class of systems that differ from each other but share a characteristic structure. The elements of such a class of systems are collectively known as an ensemble. As we introduce and study mathematical models, we should recognize that our primary objective is to represent properties of real systems. We must therefore develop an understanding of the nature of models and modeling, and how they can pertain to either simple or complex systems.

1.1 Iterative Maps (and Chaos)

An iterative map f is a function that evolves the state of a system s in discrete time

$$s(t) = f(s(t - \delta t)) \quad (1.1.1)$$

where $s(t)$ describes the state of the system at time t . For convenience we will generally measure time in units of δt which then has the value 1, and time takes integral values starting from the initial condition at $t = 0$.

Many of the complex systems we will consider in this text are of the form of Eq. (1.1.1), if we allow s to be a general variable of arbitrary dimension. The generality of iterative maps is discussed at the end of this section. We start by considering several examples of iterative maps where s is a single variable. We discuss briefly the binary variable case, $s = \pm 1$. Then we discuss in greater detail two types of maps with s a real variable, $s \in \mathbb{R}$, linear maps and quadratic maps. The quadratic iterative map is a simple model that can display complex dynamics. We assume that an iterative map may be started at any initial condition allowed by a specified domain of its system variable.

1.1.1 Binary iterative maps

There are only a few binary iterative maps. Question 1.1.1 is a complete enumeration of them.*

Question 1.1.1 Enumerate all possible iterative maps where the system is described by a single binary variable, $s = \pm 1$.

Solution 1.1.1 There are only four possibilities:

$$\begin{aligned} s(t) &= 1 \\ s(t) &= -1 \\ s(t) &= s(t - 1) \\ s(t) &= -s(t - 1) \end{aligned} \quad (1.1.2)$$

*Questions are an integral part of the text. They are designed to promote independent thought. The reader is encouraged to read the question, contemplate or work out an answer and then read the solution provided in the text. The continuation of the text assumes that solutions to questions have been read.

It is instructive to consider these possibilities in some detail. The main reason there are so few possibilities is that the form of the iterative map we are using depends, at most, on the value of the system in the previous time. The first two examples are constants and don't even depend on the value of the system at the previous time. The third map can only be distinguished from the first two by observation of its behavior when presented with two different initial conditions.

The last of the four maps is the only map that has any sustained dynamics. It cycles between two values in perpetuity. We can think about this as representing an oscillator. ■

Question 1.1.2

- In what way can the map $s(t) = -s(t-1)$ represent a physical oscillator?
- How can we think of the static map, $s(t) = s(t-1)$, as an oscillator?
- Can we do the same for the constant maps $s(t) = 1$ and $s(t) = -1$?

Solution 1.1.2 (a) By looking at the oscillator displacement with a strobe at half-cycle intervals, our measured values can be represented by this map. (b) By looking at an oscillator with a strobe at cycle intervals. (c) You might think we could, by picking a definite starting phase of the strobe with respect to the oscillator. However, the constant map ignores the first value, the oscillator does not. ■

1.1.2 Linear iterative maps: free motion, oscillation, decay and growth

The simplest example of an iterative map with s real, $s \in \mathfrak{R}$, is a constant map:

$$s(t) = s_0 \quad (1.1.3)$$

No matter what the initial value, this system always takes the particular value s_0 . The constant map may seem trivial, however it will be useful to compare the constant map with the next class of maps.

A linear iterative map with unit coefficient is a model of free motion or propagation in space:

$$s(t) = s(t-1) + v \quad (1.1.4)$$

at successive times the values of s are separated by v , which plays the role of the velocity.

Question 1.1.3 Consider the case of zero velocity

$$s(t) = s(t-1) \quad (1.1.5)$$

How is this different from the constant map?

Solution 1.1.3 The two maps differ in their dependence on the initial value. ■

Runaway growth or decay is a multiplicative iterative map:

$$s(t) = gs(t-1) \quad (1.1.6)$$

We can generate the values of this iterative map at all times by using the equivalent expression

$$s(t) = g^t s_0 = e^{\ln(g)t} s_0 \quad (1.1.7)$$

which is exponential growth or decay. The iterative map can be thought of as a sequence of snapshots of Eq. (1.1.7) at integral time. $g = 1$ reduces this map to the previous case.

Question 1.1.4 We have seen the case of free motion, and now jumped to the case of growth. What happened to accelerated motion? Usually we would consider accelerated motion as the next step after motion with a constant velocity. How can we write accelerated motion as an iterative map?

Solution 1.1.4 The description of accelerated motion requires two variables: position and velocity. The iterative map would look like:

$$\begin{aligned} x(t) &= x(t-1) + v(t-1) \\ v(t) &= v(t-1) + a \end{aligned} \quad (1.1.8)$$

This is a two-variable iterative map. To write this in the notation of Eq. (1.1.1) we would define s as a vector $s(t) = (x(t), v(t))$. ■

Question 1.1.5 What happens in the rightmost exponential expression in Eq. (1.1.7) when g is negative?

Solution 1.1.5 The logarithm of a negative number results in a phase $i\pi$. The term $i\pi t$ in the exponent alternates sign every time step as one would expect from Eq. (1.1.6). ■

At this point, it is convenient to introduce two graphical methods for describing an iterative map. The first is the usual way of plotting the value of s as a function of time. This is shown in the left panels of Fig. 1.1.1. The second type of plot, shown in the right panels, has a different purpose. This is a plot of the iterative relation $s(t)$ as a function of $s(t-1)$. On the same axis we also draw the line for the identity map $s(t) = s(t-1)$. These two plots enable us to graphically obtain the successive values of s as follows. Pick a starting value of s , which we can call $s(0)$. Mark this value on the abscissa. Mark the point on the graph of $s(t)$ that corresponds to the point whose abscissa is $s(0)$, i.e., the point $(s(0), s(1))$. Draw a horizontal line to intersect the identity map. The intersection point is $(s(1), s(1))$. Draw a vertical line back to the iterative map. This is the point $(s(1), s(2))$. Successive values of $s(t)$ are obtained by iterating this graphical procedure. A few examples are plotted in the right panels of Fig. 1.1.1.

In order to discuss the iterative maps it is helpful to recognize several features of these maps. First, intersection points of the identity map and the iterative map are the fixed points of the iterative map:

$$s_0 = f(s_0) \quad (1.1.9)$$

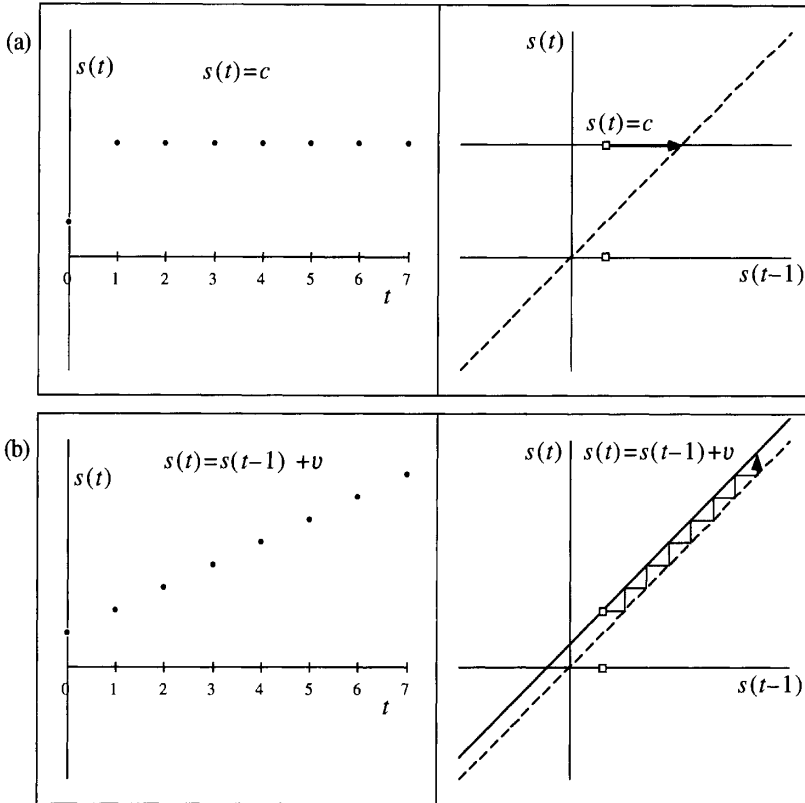
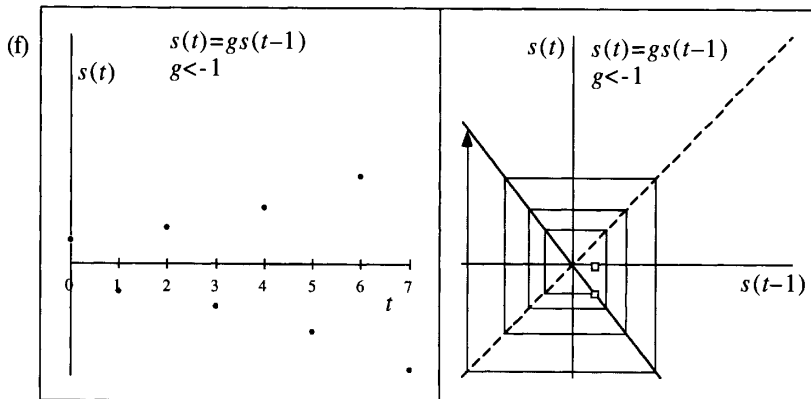
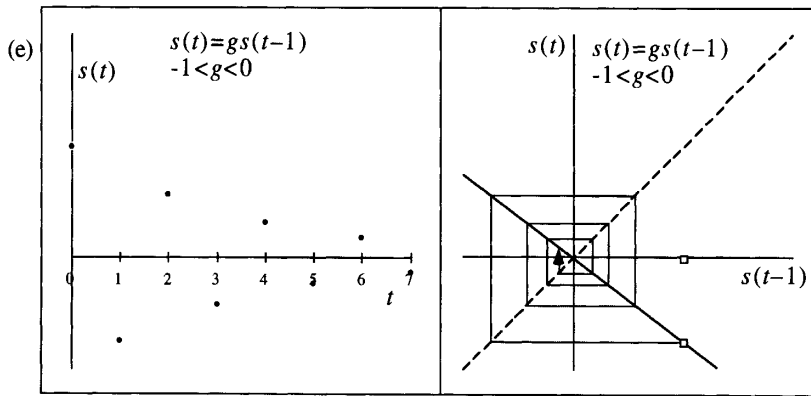
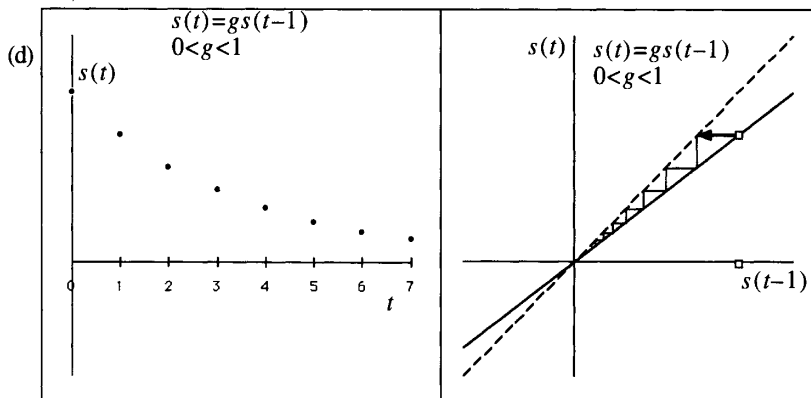
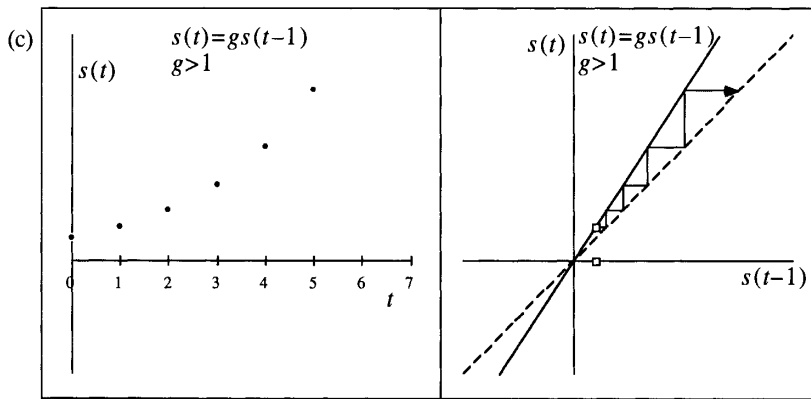


Figure 1.1.1 The left panels show the time-dependent value of the system variable $s(t)$ resulting from iterative maps. The first panel (a) shows the result of iterating the constant map; (b) shows the result of adding v to the previous value during each time interval; (c)–(f) show the result of multiplying by a constant g , where each figure shows the behavior for a different range of g values: (c) $g > 1$, (d) $0 < g < 1$, (e) $-1 < g < 0$, and (f) $g < -1$. The right panels are a different way of showing the results of iterations and are constructed as follows. First plot the function $f(s)$ (solid line), where $s(t) = f(s(t-1))$. This can be thought of as plotting $s(t)$ vs. $s(t-1)$. Second, plot the identity map $s(t) = s(t-1)$ (dashed line). Mark the initial value $s(0)$ on the horizontal axis, and the point on the graph of $s(t)$ that corresponds to the point whose abscissa is $s(0)$, i.e. the point $(s(0), s(1))$. These are shown as squares. From the point $(s(0), s(1))$ draw a horizontal line to intersect the identity map. The intersection point is $(s(1), s(1))$. Draw a vertical line back to the iterative map. This is the point $(s(1), s(2))$. Successive values of $s(t)$ are obtained by iterating this graphical procedure. ■

Fixed points, not surprisingly, play an important role in iterative maps. They help us describe the state and behavior of the system after many iterations. There are two kinds of fixed points—stable and unstable. Stable fixed points are characterized by “attracting” the result of iteration of points that are nearby. More precisely, there exists



a neighborhood of points of s_0 such that for any s in this neighborhood the sequence of points

$$\{s, f(s), f^2(s), f^3(s), \dots\} \quad (1.1.10)$$

converges to s_0 . We are using the notation $f^2(s) = f(f(s))$ for the second iteration, and similar notation for higher iterations. This sequence is just the time series of the iterative map for the initial condition s . Unstable fixed points have the opposite behavior, in that iteration causes the system to leave the neighborhood of s_0 . The two types of fixed points are also called attracting and repelling fixed points.

The family of multiplicative iterative maps in Eq. (1.1.6) all have a fixed point at $s_0 = 0$. Graphically from the figures, or analytically from Eq. (1.1.7), we see that the fixed point is stable for $|g| < 1$ and is unstable for $|g| > 1$. There is also distinct behavior of the system depending on whether g is positive or negative. For $g < 0$ the iterations alternate from one side to the other of the fixed point, whether it is attracted to or repelled from the fixed point. Specifically, if $s < s_0$ then $f(s) > s_0$ and vice versa, or $\text{sign}(s - s_0) = -\text{sign}(f(s) - s_0)$. For $g > 0$ the iteration does not alternate.

Question 1.1.6 Consider the iterative map.

$$s(t) = gs(t-1) + v \quad (1.1.11)$$

convince yourself that v does not affect the nature of the fixed point, only shifts its position.

Question 1.1.7 Consider an arbitrary iterative map of the form Eq. (1.1.1), with a fixed point s_0 (Eq. (1.1.9)). If the iterative map can be expanded in a Taylor series around s_0 show that the first derivative

$$g = \left. \frac{df(s)}{ds} \right|_{s_0} \quad (1.1.12)$$

characterizes the fixed point as follows:

For $|g| < 1$, s_0 is an attracting fixed point.

For $|g| > 1$, s_0 is a repelling fixed point.

For $g < 0$, iterations alternate sides in a sufficiently small neighborhood of s_0 .

For $g > 0$, iterations remain on one side in a sufficiently small neighborhood of s_0 .

Extra credit: Prove the same theorem for a differentiable function (no Taylor expansion needed) using the mean value theorem.

Solution 1.1.7 If the iterative map can be expanded in a Taylor series we write that

$$f(s) = f(s_0) + g(s - s_0) + h(s - s_0)^2 + \dots \quad (1.1.13)$$

where g is the first derivative at s_0 , and h is one-half of the second derivative at s_0 . Since s_0 is a fixed point $f(s_0) = s_0$ we can rewrite this as:

$$\frac{f(s) - s_0}{s - s_0} = g + h(s - s_0) + \dots \quad (1.1.14)$$

If we did not have any higher-order terms beyond g , then by inspection each of the four conditions that we have to prove would follow from this expression without restrictions on s . For example, if $|g| > 1$, then taking the magnitude of both sides shows that $f(s) - s_0$ is larger than $s - s_0$ and the iterations take the point s away from s_0 . If $g > 0$, then this expression says that $f(s)$ stays on the same side of s_0 . The other conditions follow similarly.

To generalize this argument to include the higher-order terms of the expansion, we must guarantee that whichever domain g is in ($g > 1$, $0 < g < 1$, $-1 < g < 0$, or $g < -1$), the same is also true of the whole right side. For a Taylor expansion, by choosing a small enough neighborhood $|s - s_0| < \delta$, we can guarantee the higher-order terms are less than any number ε we choose. We choose ε to be half of the minimum of $|g - 1|$, $|g - 0|$ and $|g + 1|$. Then $g + \varepsilon$ is in the same domain as g . This provides the desired guarantee and the proof is complete.

We have proven that in the vicinity of a fixed point the iterative map may be completely characterized by its first-order expansion (with the exception of the special points $g = \pm 1, 0$). ■

Thus far we have not considered the special cases $g = \pm 1, 0$. The special cases $g = 0$ and $g = 1$ have already been treated as simpler iterative maps. When $g = 0$, the fixed point at $s = 0$ is so attractive that it is the result of any iteration. When $g = 1$ all points are fixed points.

The new special case $g = -1$ has a different significance. In this case all points alternate between positive and negative values, repeating every other iteration. Such repetition is a generalization of the fixed point. Whereas in the fixed-point case we repeat every iteration, here we repeat after every two iterations. This is called a 2-cycle, and we can immediately consider the more general case of an n -cycle. In this terminology a fixed point is a 1-cycle. One way to describe an n -cycle is to say that iterating n times gives back the same result, or equivalently, that a new iterative map which is the n th fold composition of the original map $h = f^n$ has a fixed point. This description would include also fixed points of f and all points that are m -cycles, where m is a divisor of n . These are excluded from the definition of the n -cycles. While we have introduced cycles using a map where all points are 2-cycles, more general iterative maps have specific sets of points that are n -cycles. The set of points of an n -cycle is called an orbit. There are a variety of properties of fixed points and cycles that can be proven for an arbitrary map. One of these is discussed in Question 1.1.8.

Question 1.1.8 Prove that there is a fixed point between any two points of a 2-cycle if the iterating function f is continuous.

Solution 1.1.8 Let the 2-cycle be written as

$$\begin{aligned}s_2 &= f(s_1) \\ s_1 &= f(s_2)\end{aligned}\tag{1.1.15}$$

Consider the function $h(s) = f(s) - s$, $h(s_1)$ and $h(s_2)$ have opposite signs and therefore there must be an s_0 between s_1 and s_2 such that $h(s_0) = 0$ —the fixed point. ■

We can also generalize the definition of attracting and repelling fixed points to consider attracting and repelling n -cycles. Attraction and repulsion for the cycle is equivalent to the attraction and repulsion of the fixed point of f^n .

1.1.3 Quadratic iterative maps: cycles and chaos

The next iterative map we will consider describes the effect of nonlinearity (self-action):

$$s(t) = as(t-1)(1-s(t-1))\tag{1.1.16}$$

or equivalently

$$f(s) = as(1-s)\tag{1.1.17}$$

This map has played a significant role in development of the theory of dynamical systems because even though it looks quite innocent, it has a dynamical behavior that is not described in the conventional science of simple systems. Instead, Eq. (1.1.16) is the basis of significant work on chaotic behavior, and the transition of behavior from simple to chaotic. We have chosen this form of quadratic map because it simplifies somewhat the discussion. Question 1.1.11 describes the relationship between this family of quadratic maps, parameterized by a , and what might otherwise appear to be a different family of quadratic maps.

We will focus on a values in the range $4 > a > 0$. For this range, any value of s in the interval $s \in [0,1]$ stays within this interval. The minimum value $f(s) = 0$ occurs for $s = 0, 1$ and the maximal value occurs for $s = 1/2$. For all values of a there is a fixed point at $s = 0$ and there can be at most two fixed points, since a quadratic can only intersect a line (Eq. (1.1.9)) in two points.

Taking the first derivative of the iterative map gives

$$\frac{df}{ds} = a(1-2s)\tag{1.1.18}$$

At $s = 0$ the derivative is a which, by Question 1.1.7, shows that $s = 0$ is a stable fixed point for $a < 1$ and an unstable fixed point for $a > 1$. The switching of the stability of the fixed point at $s = 0$ coincides with the introduction of a second fixed point in the interval $[0,1]$ (when the slope at $s = 0$ is greater than one, $f(s) > s$ for small s , and since

$f(1) = 0$, we have that $f(s_1) = s_1$ for some s_1 in $[0,1]$ by the same construction as in Question 1.1.8). We find s_1 by solving the equation

$$s_1 = as_1(1-s_1) \quad (1.1.19)$$

$$s_1 = (a-1)/a \quad (1.1.20)$$

Substituting this into Eq. (1.1.18) gives

$$\left. \frac{df}{ds} \right|_{s_1} = 2-a \quad (1.1.21)$$

This shows that for $1 < a < 3$, the new fixed point is stable by Question 1.1.7. Moreover, the derivative is positive for $1 < a < 2$, so s_1 is stable and convergence is from one side. The derivative is negative for $2 < a < 3$, so s_1 is stable and alternating.

Fig. 1.1.2(a)–(c) shows the three cases: $a = 0.5$, $a = 1.5$ and $a = 2.8$. For $a = 0.5$, starting from anywhere within $[0,1]$ leads to convergence to $s = 0$. When $s(0) > 0.5$ the first iteration takes the system to $s(1) < 0.5$. The closer we start to $s(0) = 1$ the closer to $s = 0$ we get in the first jump. At $s(0) = 1$ the convergence to 0 occurs in the first jump. A similar behavior would be found for any value of $0 < a < 1$. For $a = 1.5$ the behavior is more complicated. Except for the points $s = 0, 1$, the convergence is always to the fixed point $s_1 = (a-1)/a$ between 0 and 1. For $a = 2.8$ the iterations converge to the same point; however, the convergence is alternating. Because there can be at most two fixed points for the quadratic map, one might think that this behavior would be all that would happen for $1 < a < 4$. One would be wrong. The first indication that this is not the case is the instability of the fixed point at s_1 starting from $a = 3$.

What happens for $a > 3$? Both of the fixed points that we have found, and the only ones that can exist for the quadratic map, are now unstable. We know that the iteration of the map has to go somewhere, and only within $[0,1]$. The only possibility, within our experience, is that there is an attracting n -cycle to which the fixed points are unstable. Let us then consider the map $f^2(s)$ whose fixed points are 2-cycles of the original map. $f^2(s)$ is shown in the right panels of Fig. 1.1.2 for increasing values of a . The fixed points of $f(s)$ are also fixed points of $f^2(s)$. However, we see that two additional fixed points exist for $a > 3$. We can also show analytically that two fixed points are introduced at exactly $a = 3$:

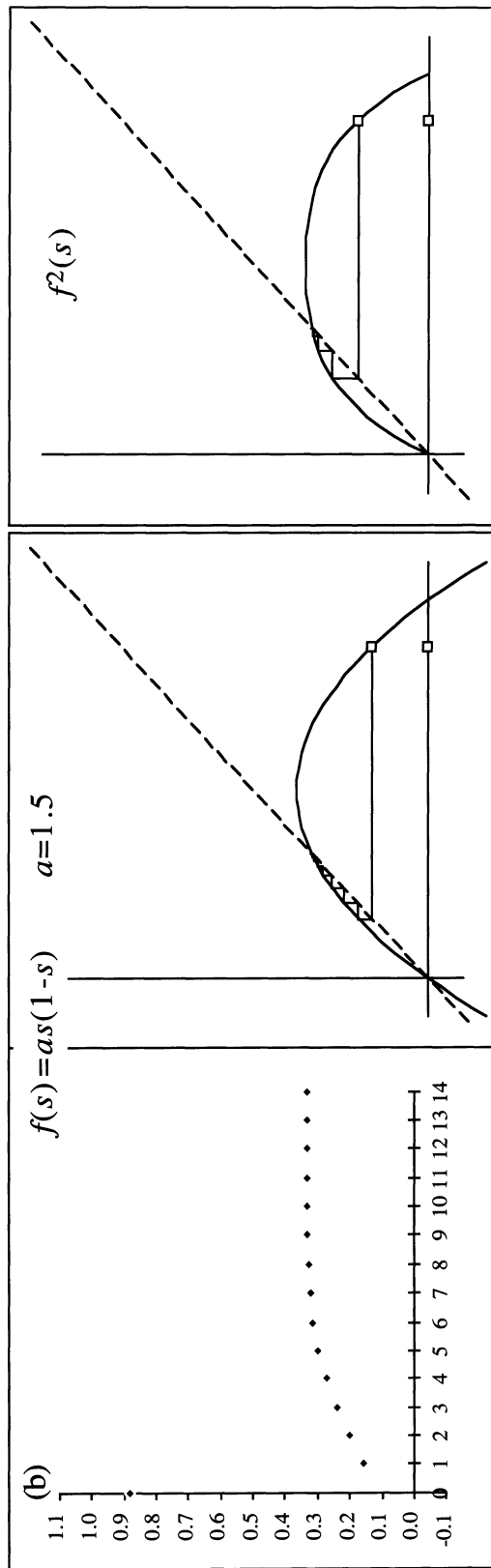
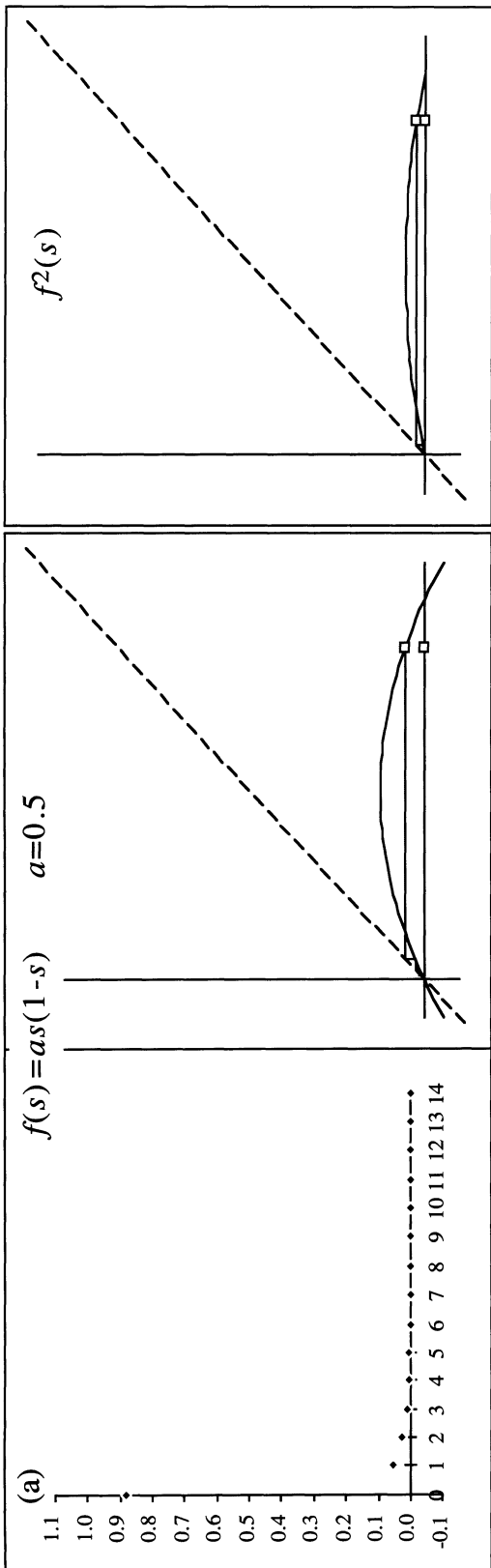
$$f^2(s) = a^2 s(1-s)(1-as(1-s)) \quad (1.1.22)$$

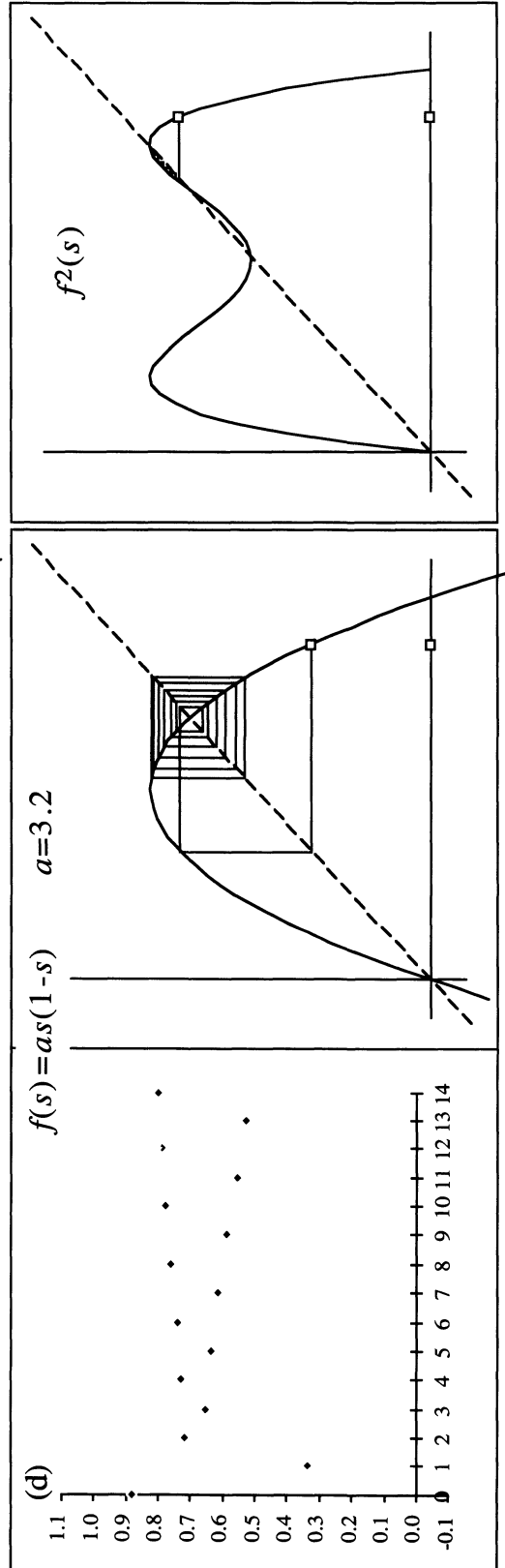
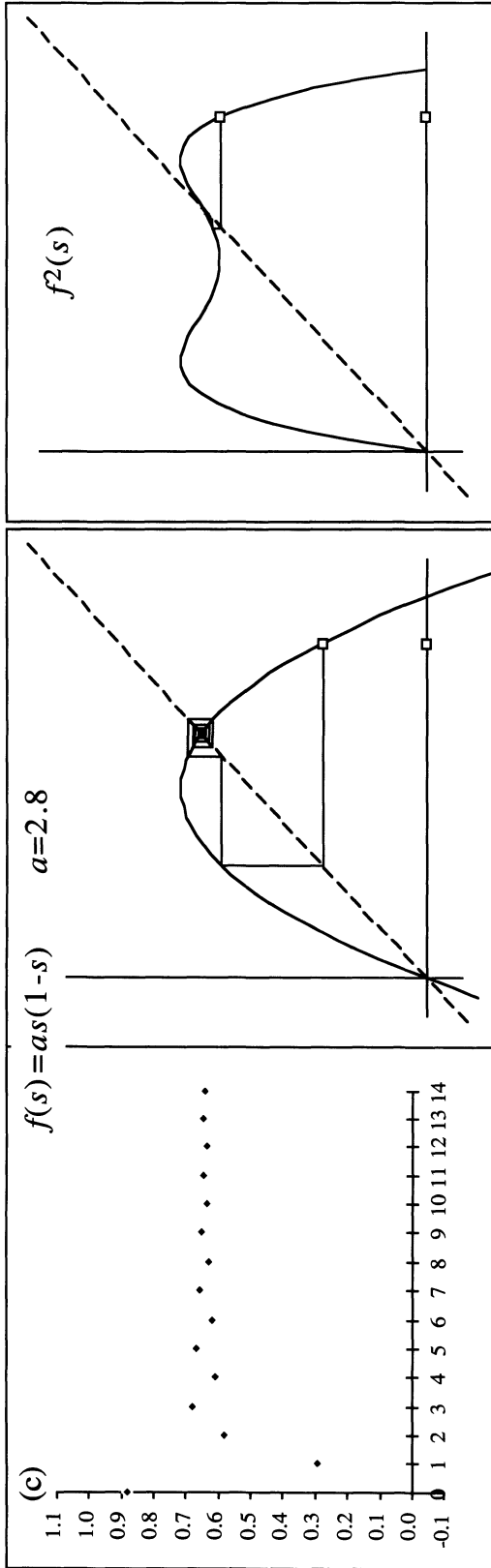
To find the fixed point we solve:

$$s = a^2 s(1-s)(1-as(1-s)) \quad (1.1.23)$$

We already know two solutions of this quartic equation—the fixed points of the map f . One of these at $s = 0$ is obvious. Dividing by s we have a cubic equation:

$$a^3 s^3 - 2a^3 s^2 + a^2(1+a)s + (1-a^2) = 0 \quad (1.1.24)$$





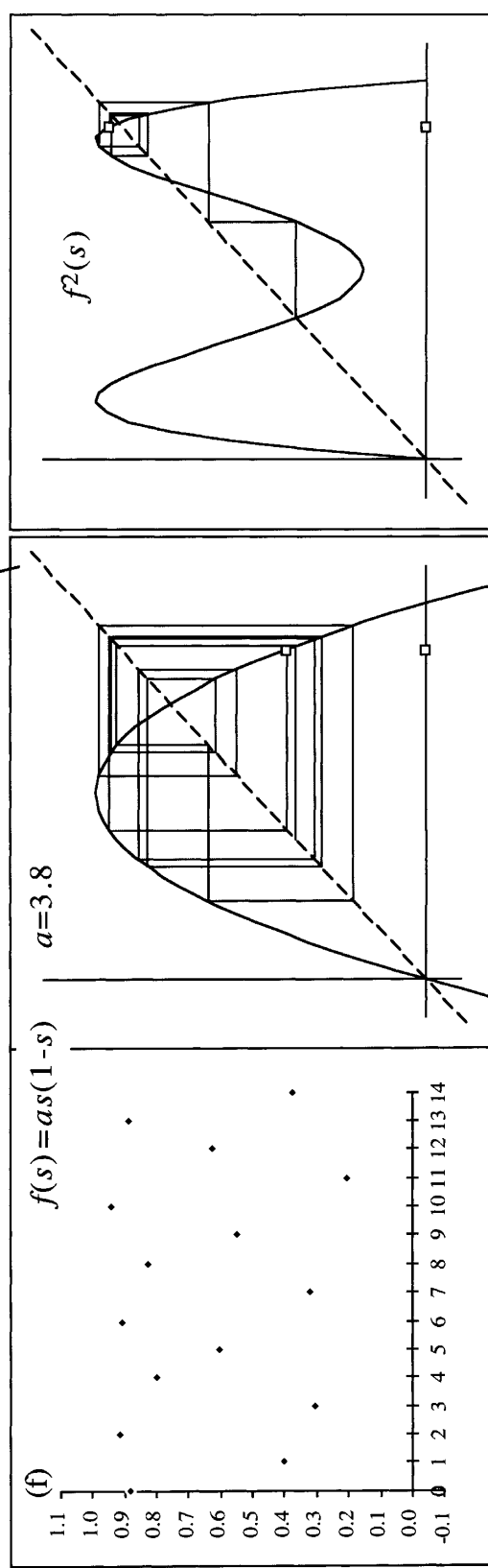
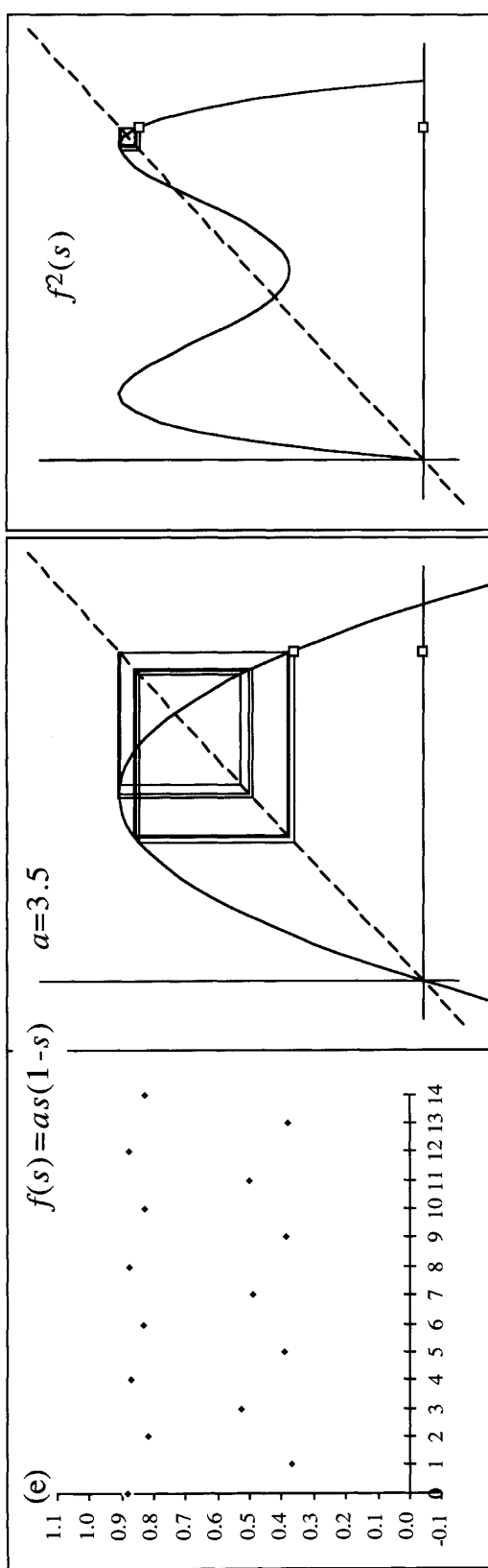


Figure 1.1.2 (pp. 28-30) Plots of the result of iterating the quadratic map $f(s) = as(1 - s)$ for different values of a . The left and center panels are similar to the left and right panels of Fig. 1.1.1. The left panels plot $s(t)$. The center panels describe the iteration of the map $f(s)$ on axes corresponding to $s(t)$ and $s(t - 1)$. The right panels are similar to the center panels but are for the function $f^2(s)$. The different values of a are indicated on the panels and show the changes from (a) convergence to $s = 0$ for $a = 0.5$, (b) convergence to $s = (a - 1)/a$ for $a = 1.5$, (c) alternating convergence to $s = (a - 1)/a$ for $a = 2.8$, (d) bifurcation — convergence to a 2-cycle for $a = 3.2$, (e) second bifurcation — convergence to a 4-cycle for $a = 3.5$, (f) chaotic behavior for $a = 3.8$. ■

We can reduce the equation to a quadratic by dividing by $(s - s_1)$ as follows (we simplify the algebra by dividing by $a(s - s_1) = (as - (a - 1))$):

$$(as - (a - 1)) \left(\frac{a^2 s^2 - a(a + 1)s + (a + 1)}{a^3 s^3 - 2a^3 s^2 + a^2(1 + a)s + (1 - a^2)} \right) = \frac{a^3 s^3 - (a - 1)a^2 s^2}{-(a + 1)a^2 s^2 + a^2(1 + a)s + (1 - a^2)} - \frac{-(a + 1)a^2 s^2 + a(1 + a)(a - 1)s}{a(1 + a)s + (1 - a^2)} \quad (1.1.25)$$

Now we can obtain the roots to the quadratic:

$$a^2 s^2 - a(a + 1)s + (a + 1) = 0 \quad (1.1.26)$$

$$s_2 = \frac{(a + 1) \pm \sqrt{(a + 1)(a - 3)}}{2a} \quad (1.1.27)$$

This has two solutions (as it must for a 2-cycle) for $a < -1$ or for $a > 3$. The former case is not of interest to us since we have assumed $0 < a < 4$. The latter case is the two roots that are promised. Notice that for exactly $a = 3$ the two roots that are the new 2-cycle are the same as the fixed point we have already found s_1 . The 2-cycle splits off from the fixed point at $a = 3$ when the fixed point becomes unstable. The two attracting points continue to separate as a increases. For $a > 3$ we expect that the result of iteration eventually settles down to the 2-cycle. The system state alternates between the two roots Eq. (1.1.27). This is shown in Fig. 1.1.2(d).

As we continue to increase a beyond 3, the 2-cycle will itself become unstable at a point that can be calculated by setting

$$\left. \frac{df^2}{ds} \right|_{s_2} = -1 \quad (1.1.28)$$

to be $a = 1 + \sqrt{6} = 3.44949$. At this value of a the 2-cycle splits into a 4-cycle (Fig. 1.1.2(e)). Each of the fixed points of $f^2(s)$ simultaneously split into 2-cycles that together form a 4-cycle for the original map.

Question 1.1.9 Show that when f has a 2-cycle, both of the fixed points of f^2 must split simultaneously.

Solution 1.1.9 The split occurs when the fixed points become unstable—the derivative of f^2 equals -1 . We can show that the derivative is equal at the two fixed points of Eq. (1.1.27), which we call $s_2^\pm$:

$$\left. \frac{df^2}{ds} \right|_{s_2} = \left. \frac{df(f(s))}{ds} \right|_{s_2} = \left. \frac{df(s)}{ds} \right|_{f(s_2)} \left. \frac{df(s)}{ds} \right|_{s_2} \quad (1.1.29)$$

where we have made use of the chain rule. Since $f(s_2^+) = s_2^-$ and vice versa, we have shown this expression is the same whether $s_2 = s_2^+$ or $s_2 = s_2^-$.

Note: This can be generalized to show that the derivative of f^k is the same at all of its k fixed points corresponding to a k -cycle of f . ■

The process of taking an n -cycle into a $2n$ -cycle is called bifurcation. Bifurcation continues to replace the limiting behavior of the iterative map with progressively longer cycles of length 2^k . The bifurcations can be simulated. They occur at smaller and smaller intervals and there is a limit point to the bifurcations at $a_c = 3.56994567$. Fig. 1.1.3 shows the values that are reached by the iterative map at long times—the stable cycles—as a function of $a < a_c$. We will discuss an algebraic treatment of the bifurcation regime in Section 1.10.

Beyond the bifurcation regime $a > a_c$ (Fig. 1.1.2(f)) the behavior of the iterative map can no longer be described using simple cycles that attract the iterations. The behavior in this regime has been identified with chaos. Chaos has been characterized in many ways, but one property is quite generally agreed upon—the inherent lack of predictability of the system dynamics. This is often expressed more precisely by describing the sensitivity of the system's fate to the initial conditions. A possible definition is: There exists a distance d such that for any neighborhood V of any point s it is possible to find a point s' within the neighborhood and a number of iterations k so that $f^k(s')$ is further than d away from $f^k(s)$. This means that arbitrarily close to any point is a point that will be displaced a significant distance away by iteration. Qualitatively, there are two missing aspects of this definition, first that the points that move far away must not be too unlikely (otherwise the system is essentially predictable) and second that d is not too small (in which case the divergence of the dynamics may not be significant).

If we look at the definition of chaotic behavior, we see that the concept of scale plays an important role. A small distance between s and s' turns into a large distance between $f^k(s)$ and $f^k(s')$. Thus a fine-scale difference eventually becomes a large-scale difference. This is the essence of chaos as a model of complex system behavior. To understand it more fully, we can think about the state variable s not as one real variable,

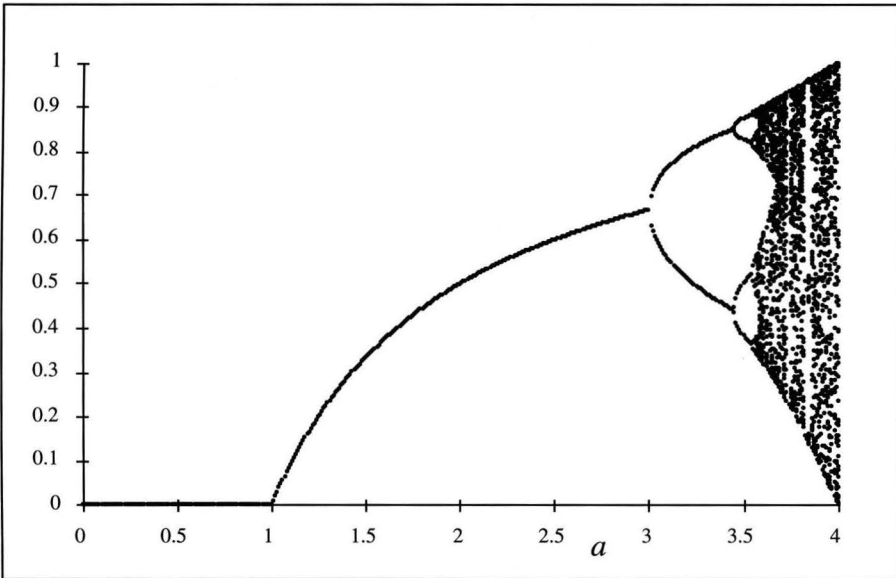


Figure 1.1.3 A plot of values of s visited by the quadratic map $f(s) = as(1 - s)$ after many iterations as a function of a , including stable points, cycles and chaotic behavior. The different regimes are readily apparent. For $a < 1$ the stable point is $s = 0$. For $1 < a < 3$ the stable point is at $s_0 = (a - 1)/a$. For $3 < a < a_c$ with $a_c = 3.56994567$, there is a bifurcation cascade with 2-cycles then 4-cycles, etc. 2^k -cycles for all values of k appear in progressively narrower regions of a . Beyond 4-cycles they cannot be seen in this plot. For $a > a_c$ there is chaotic behavior. There are regions of s values that are not visited and regions that are visited in the long time behavior of the quadratic map in the chaotic regime which this figure does not fully illustrate. ■

but as an infinite sequence of binary variables that form its binary representation $s = 0.r_1r_2r_3r_4\dots$. Each of these binary variables represents the state of the system—the value of some quantity we can measure about the system—on a particular length scale. The higher order bits represent the larger scales and the lower order ones represent the finer scales. Chaotic behavior implies that the state of the first few binary variables, r_1r_2 , at a particular time are determined by the value of fine scale variables at an earlier time. The farther back in time we look, the finer scale variables we have to consider in order to know the present values of r_1r_2 . Because many different variables are relevant to the behavior of the system, we say that the system has a complex behavior. We will return to these issues in Chapter 8.

The influence of fine length scales on coarse ones makes iterative maps difficult to simulate by computer. Computer representations of real numbers always have finite precision. This must be taken into account if simulations of iterative maps or chaotic complex systems are performed.

Another significant point about the iterative map as a model of a complex system is that there is nothing outside of the system that is influencing it. All of the information we need to describe the behavior is contained in the precise value of s . The complex behavior arises from the way the different parts of the system—the fine and course scales—affect each other.

Question 1.1.10: Why isn't the iterative map in the chaotic regime equivalent to picking a number at random?

Solution 1.1.10: We can still predict the behavior of the iterative map over a few iterations. It is only when we iterate long enough that the map becomes unpredictable. More specifically, the continuity of the function $f(s)$ guarantees that for s and s' close together $f(s)$ and $f(s')$ will also be close together. Specifically, given an ε it is possible to find a δ such that for $|s - s'| < \delta$, $|f(s) - f(s')| < \varepsilon$. For the family of functions we have been considering, we only need to set $\delta < \varepsilon/a$, since then we have:

$$|f(s) - f(s')| = a|s(1-s) - s'(1-s')| = a|s - s'| |1 - (s + s')| < a|s - s'| < \varepsilon \quad (1.1.30)$$

Thus if we fix the number of cycles to be k , we can always find two points close enough so that $|f^k(s') - f^k(s)| < \varepsilon$ by setting $|s - s'| < \varepsilon/a^k$. ■

The tuning of the parameter a leading from simple convergent behavior through cycle bifurcation to chaos has been identified as a universal description of the appearance of chaotic behavior from simple behavior of many systems. How do we take a complicated real system and map it onto a discrete time iterative map? We must define a system variable and then take snapshots of it at fixed intervals (or at least well-defined intervals). The snapshots correspond to an iterative map. Often there is a natural choice for the interval that simplifies the iterative behavior. We can then check to see if there is bifurcation and chaos in the real system when parameters that control the system behavior are varied.

One of the earliest examples of the application of iterative maps is to the study of heart attacks. Heart attacks occur in many different ways. One kind of heart attack is known as fibrillation. Fibrillation is characterized by chaotic and ineffective heart muscle contractions. It has been suggested that bifurcation may be observed in heartbeats as a period doubling (two heartbeats that are inequivalent). If correct, this may serve as a warning that the heart structure, due to various changes in heart tissue parameters, may be approaching fibrillation. Another system where more detailed studies have suggested that bifurcation occurs as a route to chaotic behavior is that of turbulent flows in hydrodynamic systems. A subtlety in the application of the ideas of bifurcation and chaos to physical systems is that physical systems are better modeled as having an increasing number of degrees of freedom at finer scales. This is to be contrasted with a system modeled by a single real number, which has the same number of degrees of freedom (represented by the binary variables above) at each length scale.

1.1.4 Are all dynamical systems iterative maps?

How general is the iterative map as a tool for describing the dynamics of systems? There are three apparent limitations of iterative maps that we will consider modifying later, Eq. (1.1.1):

- describes the homogeneous evolution of a system since f itself does not depend on time,
- describes a system where the state of the system at time t depends only on the state of the system at time $t - \delta t$, and
- describes a deterministic evolution of a system.

We can, however, bypass these limitations and keep the same form of the iterative map if we are willing to let s describe not just the present state of the system but also

- the state of the system and all other factors that might affect its evolution in time,
- the state of the system at the present time and sufficiently many previous times, and
- the probability that the system is in a particular state.

Taking these caveats together, all of the systems we will consider are iterative maps, which therefore appear to be quite general. Generality, however, can be quite useless, since we want to discard as much information as possible when describing a system.

Another way to argue the generality of the iterative map is through the laws of classical or quantum dynamics. If we consider s to be a variable that describes the positions and velocities of all particles in a system, all closed systems described by classical mechanics can be described as deterministic iterative maps. Quantum evolution of a closed system may also be described by an iterative map if s describes the wave function of the system. However, our intent is not necessarily to describe microscopic dynamics, but rather the dynamics of variables that we consider to be relevant in describing a system. In this case we are not always guaranteed that a deterministic iterative map is sufficient. We will discuss relevant generalizations, first to stochastic maps, in Section 1.2.

Extra Credit Question 1.1.11 Show that the system of quadratic iterative maps

$$s(t) = s(t-1)^2 + k \quad (1.1.31)$$

is essentially equivalent in its dynamical properties to the iterative maps we have considered in Eq. (1.1.16).

Solution 1.1.11 Two iterative maps are equivalent in their properties if we can perform a time-independent one-to-one map of the time-dependent system states from one case to the other. We will attempt to transform the family of quadratic maps given in this problem to the one of Eq. (1.1.16) using a linear map valid at all times

$$s(t) = ms'(t) + b \quad (1.1.32)$$

By direct substitution this leads to:

$$ms'(t) + b = (ms'(t-1) + b)^2 + k \quad (1.1.33)$$

We must now choose the values of m and b so as to obtain the form of Eq. (1.1.16).

$$s'(t) = ms'(t-1)(s'(t-1) + \frac{2b}{m}) + \frac{1}{m}(b^2 + k - b) \quad (1.1.34)$$

For a correct placement of minus signs in the parenthesis we need:

$$s'(t) = (-m)s'(t-1)\left(-\frac{2b}{m}\right) - s'(t-1) + \frac{1}{m}(b^2 + k - b) \quad (1.1.35)$$

or

$$b^2 - b + k = 0 \quad (1.1.36)$$

$$\frac{2b}{m} = -1 \quad (1.1.37)$$

giving

$$b = (1 \pm \sqrt{1 - 4k})/2 \quad (1.1.38)$$

$$a = -m = 2b = (1 \pm \sqrt{1 - 4k}) \quad (1.1.39)$$

We see that for $k < 1/4$ we have two solutions. These solutions give all possible (positive and negative) values of a .

What about $k > 1/4$? It turns out that this case is not very interesting compared to the rich behavior for $k < 1/4$, since there are no finite fixed points, and therefore by Question 1.1.8 no 2-cycles (it is not hard to generalize this to n -cycles). To confirm this, verify that iterations diverge to $+\infty$ from any initial condition.

Note: The system of equations of this question are the ones extensively analyzed by Devaney in his excellent textbook *A First Course in Chaotic Dynamical Systems*. ■

Extra Credit Question 1.1.12 You are given a problem to solve which when reduced to mathematical form looks like

$$s = f_c(s) \quad (1.1.40)$$

where f is a complicated function that depends on a parameter c . You know that there is a solution of this equation in the vicinity of s_0 . To solve this equation you try to iterate it (Newton's method) and it works, since you find that $f^k(s_0)$ converges nicely to a solution. Now, however, you realize that you need to solve this problem for a slightly different value of the parameter c , and when you try to iterate the equation you can't get the value of s to converge. Instead the values start to oscillate and then behave in a completely erratic

way. Suggest a solution for this problem and see if it works for the function $f_c(s) = cs(1-s)$, $c = 3.8$, $s_0 = 0.5$. A solution is given in stages (a) - (c) below.

Solution 1.1.12(a) A common resolution of this problem is to consider iterating the function:

$$h_c(s) = \alpha s + (1-\alpha)f_c(s) \quad (1.1.41)$$

where we can adjust α to obtain rapid convergence. Note that solutions of

$$s = h_c(s) \quad (1.1.42)$$

are the same as solutions of the original problem.

Question 1.1.12(b) Explain why this could work.

Solution 1.1.12(b) The derivative of this function at a fixed point can be controlled by the value of α . It is a linear interpolation between the fixed point derivative of f_c and 1. If the fixed point is unstable and oscillating, the derivative of f_c must be less than -1 and the interpolation should help.

We can also explain this result without appealing to our work on iterative maps by noting that if the iteration is causing us to overshoot the mark, it makes sense to mix the value s we start from with the value we get from $f_c(s)$ to get a better estimate.

Question 1.1.12(c) Explain how to pick α .

Solution 1.1.12(c) If the solution is oscillating, then it makes sense to assume that the fixed point is in between successive values and the distance is revealed by how much further it gets each time; i.e., we assume that the iteration is essentially a linear map near the fixed point and we adjust α so that we compensate exactly for the overshoot of f_c .

Using two trial iterations, a linear approximation to f_c at s_0 looks like:

$$\begin{aligned} s_2 &= f_c(s_1) \approx g(s_1 - s_0) + s_0 \\ s_3 &= f_c(s_2) \approx g(s_2 - s_0) + s_0 \end{aligned} \quad (1.1.43)$$

Adopting the linear approximation as a definition of g we have:

$$g \equiv (s_3 - s_2)/(s_2 - s_1) \quad (1.1.44)$$

Set up α so that the first iteration of the modified system will take you to the desired answer:

$$s_0 = \alpha s_1 + (1-\alpha)f_c(s_1) \quad (1.1.45)$$

or

$$s_0 - s_1 = (1-\alpha)(f_c(s_1) - s_1) = (1-\alpha)(s_2 - s_1) \quad (1.1.46)$$

$$(1-\alpha) = (s_0 - s_1)/(s_2 - s_1) \quad (1.1.47)$$

To eliminate the unknown s_0 we use Eq. (1.1.43) to obtain:

$$(s_2 - s_1) = g(s_1 - s_0) + (s_0 - s_1) \quad (1.1.48)$$

$$(s_0 - s_1) = (s_2 - s_1)/(1 - g) \quad (1.1.49)$$

or

$$1 - \alpha = 1/(1 - g) \quad (1.1.50)$$

$$\alpha = -g/(1 - g) = (s_2 - s_3)/(2s_2 - s_1 - s_3) \quad (1.1.51)$$

It is easy to check, using the formula in terms of g , that the modified iteration has a zero derivative at s_0 when we use the approximate linear forms for f_c . This means we have the best convergence possible using the information from two iterations of f_c . We then use the value of α to iterate to convergence.

Try it! ■

1.2 Stochastic Iterative Maps

Many of the systems we would like to consider are described by system variables whose value at the next time step we cannot predict with complete certainty. The uncertainty may arise from many sources, including the existence of interactions and parameters that are too complicated or not very relevant to our problem. We are then faced with describing a system in which the outcome of an iteration is probabilistic and not deterministic. Such systems are called stochastic systems. There are several ways to describe such systems mathematically. One of them is to consider the outcome of a particular update to be selected from a set of possible values. The probability of each of the possible values must be specified. This description is not really a model of a single system, because each realization of the system will do something different. Instead, this is a model of a collection of systems—an ensemble. Our task is to study the properties of this ensemble.

A stochastic system is generally described by the time evolution of random variables. We begin the discussion by defining a random variable. A random variable s is defined by its probability distribution $P_s(s')$, which describes the likelihood that s has the value s' . If s is a continuous variable, then $P_s(s')ds'$ is the probability that s resides between s' and $s' + ds'$. Note that the subscript is the variable name rather than an index. For example, s might be a binary variable that can have the value $+1$ or -1 . $P_s(1)$ is the probability that $s = 1$ and $P_s(-1)$ is the probability that $s = -1$. If s is the outcome of an unbiased coin toss, with heads called 1 and tails called -1 , both of these values are $1/2$. When no confusion can arise, the notation $P_s(s')$ is abbreviated to $P(s)$, where s may be either the variable or the value. The sum over all possible values of the probability must be 1 .

$$\sum_{s'} P_s(s') = 1 \quad (1.2.1)$$

In the discussion of a system described by random variables, we often would like to know the average value of some quantity $Q(s)$ that depends in a definite way on the value of the stochastic variable s . This average is given by:

$$\langle Q(s) \rangle = \sum_{s'} P_s(s') Q(s') \quad (1.2.2)$$

Note that the average is a linear operation.

We now consider the case of a time-dependent random variable. Rather than describing the time dependence of the variable $s(t)$, we describe the time dependence of the probability distribution $P_s(s'; t)$. Similar to the iterative map, we can consider the case where the outcome only depends on the value of the system variable at a previous time, and the transition probabilities do not depend explicitly on time. Such systems are called Markov chains. The transition probabilities from a state at a particular time to the next discrete time are written:

$$P_s(s'(t) | s'(t-1)) \quad (1.2.3)$$

P_s is used as the notation for the transition probability, since it is also the probability distribution of s at time t , given a particular value $s'(t-1)$ at the previous time. The use of a time index for the arguments illustrates the use of the transition probability. $P_s(1|1)$ is the probability that when $s = 1$ at time $t-1$ then $s = 1$ at time t . $P_s(-1|1)$ is the probability that when $s = 1$ at time $t-1$ then $s = -1$ at time t . The transition probabilities, along with the initial probability distribution of the system $P_s(s'; t=0)$, determine the time-dependent ensemble that we are interested in. Assuming that we don't lose systems on the way, the transition probabilities of Eq. (1.2.3) must satisfy:

$$\sum_{s''} P_s(s'' | s') = 1 \quad (1.2.4)$$

This states that no matter what the value of the system variable is at a particular time, it must reach some value at the next time.

The stochastic system described by transition probabilities can be written as an iterative map on the probability distribution $P(s)$

$$P_s(s'; t) = \sum_{s''} P_s(s' | s'') P_s(s''; t-1) \quad (1.2.5)$$

It may be more intuitive to write this using the notation

$$P_s(s'(t); t) = \sum_{s'(t-1)} P_s(s'(t) | s'(t-1)) P_s(s'(t-1); t-1) \quad (1.2.6)$$

in which case it may be sufficient, though hazardous, to write the abbreviated form

$$P(s(t)) = \sum_{s(t-1)} P(s(t) | s(t-1)) P(s(t-1)) \quad (1.2.7)$$

It is important to recognize that the time evolution equation for the probability is linear. The linear evolution of this system (Eq. (1.2.5)) guarantees that superposition applies. If we start with an initial distribution $P(s;0) = \frac{1}{2}P^1(s;0) + \frac{1}{2}P^2(s;0)$ at time $t = 0$, then we could find the result at time t by separately looking at the evolution of each of the probabilities $P^1(s;0)$ and $P^2(s;0)$. Explicitly we can write $P(s;t) = \frac{1}{2}P^1(s;t) + \frac{1}{2}P^2(s;t)$. The meaning of this equation should be well noted. The right side of the equation is the sum of the evolved probabilities $P^1(s;t)$ and $P^2(s;t)$. This linearity is a direct consequence of the independence of different members of the ensemble and says nothing about the complexity of the dynamics.

We note that ultimately we are interested in the behavior of a particular system $s(t)$ that only has one value of s at every time t . The ensemble describes how many such systems will behave. Analytically it is easier to describe the ensemble as a whole, however, simulations may also be used to observe the behavior of a single system.

1.2.1 Random walk

Stochastic systems with only one binary variable might seem to be trivial, but we will devote quite a bit of attention to this problem. We begin by considering the simplest possible binary stochastic system. This is the system which corresponds to a coin toss. Ideally, for each toss there is equal probability of heads ($s = +1$) or tails ($s = -1$), and there is no memory from one toss to the next. The ensemble at each time is independent of time and has an equal probability of ± 1 :

$$P(s;t) = \frac{1}{2}\delta_{s,1} + \frac{1}{2}\delta_{s,-1} \quad (1.2.8)$$

where the discrete delta function is defined by

$$\delta_{i,j} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad (1.2.9)$$

Since Eq. (1.2.8) is independent of what happens at all previous times, the evolution of the state variable is given by the same expression

$$P(s'|s) = \frac{1}{2}\delta_{s',1} + \frac{1}{2}\delta_{s',-1} \quad (1.2.10)$$

We can illustrate the evaluation of the average of a function of s at time t :

$$\langle Q(s) \rangle_t = \sum_{s'=\pm 1} Q(s')P_s(s';t) = \sum_{s'=\pm 1} Q(s') \left(\frac{1}{2}\delta_{s',1} + \frac{1}{2}\delta_{s',-1} \right) = \frac{1}{2} \sum_{s'=\pm 1} Q(s') \quad (1.2.11)$$

For example, if we just take $Q(s)$ to be s itself we have the average of the system variable:

$$\langle s \rangle_t = \frac{1}{2} \sum_{s'=\pm 1} s' = 0 \quad (1.2.12)$$

Question 1.2.1 Will you win more fair coin tosses if (a) you pick heads every time, or if (b) you alternate heads and tails, or if (c) you pick heads or tails at random or if (d) you pick heads and tails by some other system? Explain why.

Solution 1.2.1 In general, we cannot predict the number of coin tosses that will be won, we can only estimate it based on the chance of winning. Assuming a fair coin means that this is the best that can be done. Any of the possibilities (a)–(c) give the same chance of winning. In none of these ways of gambling does the choice you make correlate with the result of the coin toss. The only system (d) that can help is if you have some information about what the result of the toss will be, like betting on the known result *after* the coin is tossed. A way to write this formally is to write the probability distribution of the choice that you are making. This choice is also a stochastic process. Calling the choice $c(t)$, the four possibilities mentioned are:

$$(a) \quad P(c;t) = \delta_{c,1} \quad (1.2.13)$$

$$(b) \quad P(c;t) = \frac{1+(-1)^t}{2} \delta_{c,1} + \frac{1-(-1)^t}{2} \delta_{c,-1} = \text{mod}_2(t) \delta_{c,1} + \text{mod}_2(t+1) \delta_{c,-1} \quad (1.2.14)$$

$$(c) \quad P(c;t) = \frac{1}{2} \delta_{c,1} + \frac{1}{2} \delta_{c,-1} \quad (1.2.15)$$

$$(d) \quad P(c;t) = \delta_{c,s(t)} \quad (1.2.16)$$

It is sufficient to show that the average probability of winning is the same in each of (a)–(c) and is just $1/2$. We follow through the manipulations in order to illustrate some concepts in the treatment of more than one stochastic variable. We have to sum over the probabilities of each of the possible values of the coin toss and each of the values of the choices, adding up the probability that they coincide at a particular time t :

$$\langle \delta_{c,s} \rangle = \sum_{s'} \sum_{c'} \delta_{c',s'} P_s(s';t) P_c(c';t) \quad (1.2.17)$$

This expression assumes that the values of the coin toss and the value of the choice are independent, so that the joint probability of having a particular value of s and a particular value of c is the product of the probabilities of each of the variables independently:

$$P_{s,c}(s',c';t) = P_s(s';t) P_c(c';t) \quad (1.2.18)$$

—the probabilities-of-independent-variables factor. This is valid in cases (a)–(c) and not in case (d), where the probability of c occurring is explicitly a function of the value of s .

We evaluate the probability of winning in each case (a) through (c) using

$$\begin{aligned}
\langle \delta_{c,s} \rangle &= \sum_{s'} \sum_{c'} \delta_{c',s'} \left(\frac{1}{2} \delta_{s',1} + \frac{1}{2} \delta_{s',-1} \right) P_c(c';t) \\
&= \sum_{c'} \left(\frac{1}{2} \delta_{c',1} + \frac{1}{2} \delta_{c',-1} \right) P_c(c';t) \\
&= \sum_{c'} \left(\frac{1}{2} P_c(1;t) + \frac{1}{2} P_c(-1;t) \right) = \frac{1}{2}
\end{aligned} \tag{1.2.19}$$

where the last equality follows from the normalization of the probability (the sum over all possibilities must be 1, Eq. (1.2.1)) and does not depend at all on the distribution. This shows that the independence of the variables guarantees that the probability of a win is just 1/2.

For the last case (d) the trivial answer, that a win is guaranteed by this method of gambling, can be arrived at formally by evaluating

$$\langle \delta_{c,s} \rangle = \sum_{s'} \sum_{c'} \delta_{c',s'} P_{s,c}(s',c';t) \tag{1.2.20}$$

The value of s at time t is independent of the value of c , but the value of c depends on the value of s . The joint probability $P_{s,c}(s',c';t)$ may be written as the product of the probability of a particular value of $s = s'$ times the conditional probability $P_c(c'|s';t)$ of a particular value of $c = c'$ given the assumed value of s :

$$\begin{aligned}
\langle \delta_{c,s} \rangle &= \sum_{s'} \sum_{c'} \delta_{c',s'} P_s(s';t) P_c(c'|s';t) \\
&= \sum_{s'} \sum_{c'} \delta_{c',s'} P_s(s';t) \delta_{c',s'} = \sum_{s'} P_s(s';t) = 1
\end{aligned} \tag{1.2.21} \blacksquare$$

The next step in our analysis of the binary stochastic system is to consider the behavior of the sum of $s(t)$ over a particular number of time steps. This sum is the difference between the total number of heads and the total number of tails. It is equivalent to asking how much you will win or lose if you gamble an equal amount of money on each coin toss after a certain number of bets. This problem is known as a random walk, and we will define it as a consideration of the state variable

$$d(t) = \sum_{t'=1}^t s(t') \tag{1.2.22}$$

The way to write the evolution of the state variable is:

$$P(d' | d) = \frac{1}{2} \delta_{d',d+1} + \frac{1}{2} \delta_{d',d-1} \tag{1.2.23}$$

Thus a random walk considers a state variable d that can take integer values $d \in \{ \dots, -1, 0, 1, \dots \}$. At every time step, $d(t)$ can only move to a value one higher or one lower than where it is. We assume that the probability of a step to the right (higher) is equal to that of a step to the left (lower). For convenience, we assume (with no loss of gener-

ality) that the system starts at position $d(0) = 0$. This is built into Eq. (1.2.22). Because of the symmetry of the system under a shift of the origin, this is equivalent to considering any other starting point. Once we solve for the probability distribution of d at time t , because of superposition we can also find the result of evolving any initial probability distribution $P(d; t=0)$.

We can picture the random walk as that of a drunk who has difficulty consistently moving forward. Our model of this walk assumes that the drunk is equally likely to take a step forward or backward. Starting at position 0, he moves to either $+1$ or -1 . Let's say it was $+1$. Next he moves to $+2$ or back to 0. Let's say it was 0. Next to $+1$ or -1 . Let's say it was $+1$. Next to $+2$ or 0. Let's say $+2$. Next to $+3$ or $+1$. Let's say $+1$. And so on.

What is the value of system variable $d(t)$ at time t ? This is equivalent to asking how far has the walk progressed after t steps. Of course there is no way to know how far a particular system goes without watching it. The average distance over the ensemble of systems is the average over all possible values of $s(t)$. This average is given by applying Eq. (1.2.2) or Eq. (1.2.11) to all of the variables $s(t)$:

$$\begin{aligned} \langle d(t) \rangle &= \frac{1}{2} \sum_{s(t)=\pm 1} K \frac{1}{2} \sum_{s(3)=\pm 1} \frac{1}{2} \sum_{s(2)=\pm 1} \frac{1}{2} \sum_{s(1)=\pm 1} d(t) \\ &= \sum_{t'=1}^t \langle s(t') \rangle = 0 \end{aligned} \quad (1.2.24)$$

The average is written out explicitly on the first line using Eq. (1.2.11). The second line expression can be arrived at either directly or from the linearity of the average. The final answer is clear, since it is equally likely for the walker to move to the right as to the left.

We can also ask what is a typical distance traveled by a particular walker. By typical distance we mean how far from the starting point. This can either be defined by the average absolute value of the distance, or as is more commonly accepted, the root mean square (RMS) distance:

$$\sigma(t) = \sqrt{\langle d(t)^2 \rangle} \quad (1.2.25)$$

$$\langle d(t)^2 \rangle = \left\langle \left(\sum_{t'=1}^t s(t') \right)^2 \right\rangle = \left\langle \sum_{t', t''=1}^t s(t') s(t'') \right\rangle = \sum_{t', t''=1}^t \langle s(t') s(t'') \rangle \quad (1.2.26)$$

To evaluate the average of the product of the two steps, we treat differently the case in which they are the same step and when they are different steps. When the two steps are the same one we use $s(t) = \pm 1$ to obtain:

$$\langle s(t)^2 \rangle = \langle 1 \rangle = 1 \quad (1.2.27)$$

Which follows from the normalization of the probability (or is obvious). To evaluate the average of the product of two steps at different times we need the joint probability of $s(t)$ and $s(t')$. This is the probability that each of them will take a particular

value. Because we have assumed that the steps are *independent*, the joint probability is the product of the probabilities for each one separately:

$$P(s(t), s(t')) = P(s(t))P(s(t')) \quad t \neq t' \quad (1.2.28)$$

so that for example there is 1/4 chance that $s(t) = +1$ and $s(t) = -1$. The independence of the two steps leads the average of the product of the two steps to factor:

$$\begin{aligned} \langle s(t)s(t') \rangle &= \sum_{s(t), s(t')} P(s(t), s(t')) s(t)s(t') \\ &= \sum_{s(t), s(t')} P(s(t))P(s(t')) s(t)s(t') \quad t \neq t' \\ &= \langle s(t) \rangle \langle s(t') \rangle = 0 \end{aligned} \quad (1.2.29)$$

This is zero, since either of the averages are zero. We have the combined result:

$$\langle s(t)s(t') \rangle = \delta_{t,t'} \quad (1.2.30)$$

and finally:

$$\langle d(t)^2 \rangle = \sum_{t', t''=1}^t \langle s(t')s(t'') \rangle = \sum_{t', t''=1}^t \delta_{t', t''} = \sum_{t'=1}^t 1 = t \quad (1.2.31)$$

This gives the classic and important result that a random walk travels a typical distance that grows as the square root of the number of steps taken: $\sigma(t) = \sqrt{t}$.

We can now consider more completely the probability distribution of the position of the walker at time t . The probability distribution at $t = 0$ may be written:

$$P(d; 0) = \delta_{d,0} \quad (1.2.32)$$

After the first time step the probability distribution changes to

$$P(d; 1) = \frac{1}{2}\delta_{d,1} + \frac{1}{2}\delta_{d,-1} \quad (1.2.33)$$

this results from the definition $d(1) = s(1)$. After the second step $d(2) = s(1) + s(2)$ it is:

$$P(d; 2) = \frac{1}{4}\delta_{d,2} + \frac{1}{2}\delta_{d,0} + \frac{1}{4}\delta_{d,-2} \quad (1.2.34)$$

More generally it is not difficult to see that the probabilities are given by normalized binomial coefficients, since the number of ones chosen out of t steps is equivalent to the number of powers of x in $(1+x)^t$. To reach a position d after t steps we must take $(t+d)/2$ steps to the right and $(t-d)/2$ steps to the left. The sum of these is the number of steps t and their difference is d . Since each choice has 1/2 probability we have:

$$P(d, t) = \frac{1}{2^t} \binom{t}{(d+t)/2} \delta_{t,d}^{\text{oddeven}} = \frac{1}{2^t} \frac{t!}{[(d+t)/2]! [(t-d)/2]!} \delta_{t,d}^{\text{oddeven}} \quad (1.2.35)$$

$$\delta_{t,d}^{\text{oddeven}} = \frac{(1+(-1)^{t+d})}{2}$$

where the unusual delta function imposes the condition that d takes only odd or only even values depending on whether t is odd or even.

Let us now consider what happens after a long time. The probability distribution spreads out, and a single step is a small distance compared to the typical distance traveled. We can consider s and t to be continuous variables where both conditions $d, t \gg 1$ are satisfied. Moreover, we can also consider $|d| \ll t$, because the chance that all steps will be taken in one direction becomes very small. This enables us to use Sterling's approximation to the factorial

$$x! \sim \sqrt{2\pi x} e^{-x} x^x$$

$$\ln(x!) \sim x(\ln x - 1) + \ln(\sqrt{2\pi x}) \quad (1.2.36)$$

For large t it also makes sense not to restrict d to be either odd or even. In order to allow both, we, in effect, interpolate and then take only half of the probability we have in Eq. (1.2.35). This leads to the expression:

$$P(d, t) = \frac{\sqrt{t}}{\sqrt{2\pi(t-d)(t+d)} 2^t [(d+t)/2]^{[(d+t)/2]} [(t-d)/2]^{[(t-d)/2]} e^{-(d+t)/2 - (t-d)/2}}$$

$$= \frac{(2\pi t(1-x^2))^{-1/2}}{(1+x)^{[(1+x)t/2]} (1-x)^{[(1-x)t/2]}} \quad (1.2.37)$$

where we have defined $x = d/t$. To approximate this expression it is easier to consider it in logarithmic form:

$$\ln(P(d, t)) = -(t/2)[(1+x)\ln(1+x) + (1-x)\ln(1-x)] - (1/2)\ln(2\pi t(1-x^2))$$

$$\approx -(t/2)[(1+x)(x - x^2/2 + K) + (1-x)(-x - x^2/2 + K)] - (1/2)\ln(2\pi t + K)$$

$$= -tx^2/2 - \ln(\sqrt{2\pi t}) \quad (1.2.38)$$

or exponentiating:

$$P(d, t) = \frac{1}{\sqrt{2\pi t}} e^{-d^2/2t} = \frac{1}{\sqrt{2\pi \sigma}} e^{-d^2/2\sigma^2} \quad (1.2.39)$$

The prefactor of the exponential, $1/\sqrt{2\pi}\sigma$, originates from the factor $\sqrt{2\pi x}$ in Eq. (1.2.36). It is independent of d and takes care of the normalization of the probability. The result is a Gaussian distribution. Questions 1.2.2–1.2.5 investigate higher-order corrections to the Gaussian distribution.

Question 1.2.2 In order to obtain a correction to the Gaussian distribution we must add a correction term to Sterling's approximation:

$$\begin{aligned} x! &\sim \sqrt{2\pi x} e^{-x} x^x \left(1 + \frac{1}{12x} + \dots\right) \\ \ln(x!) &\sim x(\ln x - 1) + \ln(\sqrt{2\pi x}) + \ln\left(1 + \frac{1}{12x} + \dots\right) \end{aligned} \quad (1.2.40)$$

Using this expression, find the first correction term to Eq. (1.2.37).

Solution 1.2.2 The correction term in Sterling's approximation contributes a factor to Eq. (1.2.37) which is (for convenience we write here $c = 1/12$):

$$\frac{(1+c/t)}{(1+2c/(t+d))(1+2c/(t-d))} = \left(1 - \frac{3c}{t} + \dots\right) = \left(1 - \frac{1}{4t} + \dots\right) \quad (1.2.41)$$

where we have only kept the largest correction term, neglecting d compared to t . Note that the correction term vanishes as t becomes large. ■

Question 1.2.3 Keeping additional terms of the expansion in Eq. (1.2.38), and the result of Question 1.2.2, find the first order correction terms to the Gaussian distribution.

Solution 1.2.3 Correction terms in Eq. (1.2.38) arise from several places. We want to keep all terms that are of order $1/t$. To do this we must keep in mind that a typical distance traveled is $d \sim \sqrt{t}$, so that $x \sim 1/\sqrt{t}$. The next terms are obtained from:

$$\begin{aligned} \ln(P(d,t)) &= -(t/2)[(1+x)\ln(1+x) + (1-x)\ln(1-x)] \\ &\quad - (1/2)\ln(2\pi t(1-x^2)) + \ln(1-1/4t) \\ &\approx -(t/2)\left[(1+x)\left(x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 K\right) \right. \\ &\quad \left. + (1-x)\left(-x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \frac{1}{4}x^4 K\right)\right] \\ &\quad - \ln(\sqrt{2\pi t}) - (1/2)\ln(1-x^2) + \ln(1-1/4t) \\ &\approx -(t/2)\left[(x+x^2 - \frac{1}{2}x^2 - \frac{1}{2}x^3 + \frac{1}{3}x^3 + \frac{1}{3}x^4 - \frac{1}{4}x^4 K) \right. \\ &\quad \left. + (-x+x^2 - \frac{1}{2}x^2 + \frac{1}{2}x^3 - \frac{1}{3}x^3 + \frac{1}{3}x^4 - \frac{1}{4}x^4 K)\right] \\ &\quad - \ln(\sqrt{2\pi t}) + (x^2/2 + \dots) + (-1/4t + \dots) \\ &= -tx^2/2 - \ln(\sqrt{2\pi t}) - tx^4/12 + x^2/2 - 1/4t \end{aligned} \quad (1.2.42)$$

This gives us a distribution:

$$P(d, t) = \sqrt{\frac{1}{2\pi t}} e^{-d^2/2t} e^{-d^4/12t^3 + d^2/2t^2 - 1/4t} \quad (1.2.43) \blacksquare$$

Question 1.2.4 What is the size of the additional factor? Estimate the size of this term as t becomes large.

Solution 1.2.4 The typical value of the variable d is its root mean square value $\sigma = \sqrt{t}$. At this value the additional term gives a factor

$$e^{1/6t} \quad (1.2.44)$$

which approaches 1 as time increases. $\blacksquare$

Question 1.2.5 What is the fraction error that we will make if we neglect this term after one hundred steps? After ten thousand steps?

Solution 1.2.5 After one hundred time steps the walker has traveled a typical distance of ten steps. We generally approximate the probability of arriving at this distance using Eq. (1.2.39). The fractional error in the probability of arriving at this distance according to Eq. (1.2.44) is $1 - e^{1/6t} \approx -1/6t = -0.00167$. So already at a distance of ten steps the error is less than 0.2%.

It is much less likely for the walker to arrive at the distance $2\sigma = 20$. The ratio of the probability to arrive at 20 compared to 10 is $e^{-2}/e^{-0.5} \sim 0.22$. If we want to know the error of this smaller probability case we would write $(1 - e^{-16/12t + 4/2t - 1/4t}) = (1 - e^{5/12t}) \approx -0.0042$, which is a larger but still small error.

After ten thousand steps the errors are smaller than the errors at one hundred steps by a factor of one hundred. $\blacksquare$

1.2.2 Generalized random walk and the central limit theorem

We can generalize the random walk by allowing a variety of steps from the current location of the walker to sites nearby, not only to the adjacent sites and not only to integer locations. If we restrict ourselves to steps that on average are balanced left and right and are not too long ranged, we can show that all such systems have the same behavior as the simplest random walk at long enough times (and characteristically not even for very long times). This is the content of the central limit theorem. It says that summing any set of independent random variables eventually leads to a Gaussian distribution of probabilities, which is the same distribution as the one we arrived at for the random walk. The reason that the same distribution arises is that successive iteration of the probability update equation, Eq. (1.2.7), smoothes out the distribution, and the only relevant information that survives is the width of the distribution which is given by $\sigma(t)$. The proof given below makes use of a Fourier transform and can be skipped by readers who are not well acquainted with transforms. In the next section we will also include a bias in the random walk. For long times this can be described as

an average motion superimposed on the unbiased random walk. We start with the unbiased random walk.

Each step of the random walk is described by the state variable $s(t)$ at time t . The probability of a particular step size is an unspecified function that is independent of time:

$$P(s;t) = f(s) \quad (1.2.45)$$

We treat the case of integer values of s . The continuum case is Question 1.2.6. The absence of bias in the random walk is described by setting the average displacement in a single step to zero:

$$\langle s \rangle = \sum_s s f(s) = 0 \quad (1.2.46)$$

The statement above that each step is not too long ranged, is mathematically just that the mean square displacement in a single step has a well-defined value (i.e., is not infinite):

$$\langle s^2 \rangle = \sum_s s^2 f(s) = \sigma_0^2 \quad (1.2.47)$$

Eqs. (1.2.45)–(1.2.47) hold at all times.

We can still evaluate the average of $d(t)$ and the RMS value of $d(t)$ directly using the linearity of the average:

$$\langle d(t) \rangle = \left\langle \sum_{t'=1}^t s(t') \right\rangle = t \langle s \rangle = 0 \quad (1.2.48)$$

$$\langle d(t)^2 \rangle = \left\langle \left(\sum_{t'=1}^t s(t') \right)^2 \right\rangle = \sum_{t', t''=1}^t \langle s(t') s(t'') \rangle \quad (1.2.49)$$

Since $s(t')$ and $s(t'')$ are independent for $t' \neq t''$, as in Eq. (1.2.29), the average factors:

$$\langle s(t') s(t'') \rangle = \langle s(t') \rangle \langle s(t'') \rangle = 0 \quad t' \neq t'' \quad (1.2.50)$$

Thus, all terms $t' \neq t''$ are zero by Eq. (1.2.46). We have:

$$\langle d(t)^2 \rangle = \sum_{t'=1}^t \langle s(t')^2 \rangle = t \sigma_0^2 \quad (1.2.51)$$

This means that the typical value of $d(t)$ is $\sigma_0 \sqrt{t}$.

To obtain the full distribution of the random walk state variable $d(t)$ we have to sum the stochastic variables $s(t)$. Since $d(t) = d(t-1) + s(t)$ the probability of transition from $d(t-1)$ to $d(t)$ is $f(d(t) - d(t-1))$ or:

$$P(d' | d) = f(d' - d) \quad (1.2.52)$$

We can now write the time evolution equation and iterate it t times to get $P(d; t)$.

$$P(d; t) = \sum_{d'} P(d | d') P(d'; t-1) = \sum_{d'} f(d - d') P(d'; t-1) \quad (1.2.53)$$

This is a convolution, so the most convenient way to effect a t fold iteration is in Fourier space. The Fourier representation of the probability and transition functions for integral d is:

$$\begin{aligned} \tilde{P}(k; t) &\equiv \sum_d e^{-ikd} P(d; t) \\ \tilde{f}(k) &\equiv \sum_s e^{-iks} f(s) \end{aligned} \quad (1.2.54)$$

We use a Fourier series because of the restriction to integer values of d . Once we solve the problem using the Fourier representation, the probability distribution is recovered from the inverse formula:

$$P(d; t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} dk e^{ikd} \tilde{P}(k; t) \quad (1.2.55)$$

which is proved

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} dk e^{ikd} \tilde{P}(k; t) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} dk e^{ikd} \sum_{d'} e^{-ikd'} P(d'; t) \\ &= \frac{1}{2\pi} \sum_{d'} P(d'; t) \int_{-\pi}^{\pi} dk e^{ik(d-d')} = \sum_{d'} P(d'; t) \delta_{d, d'} = P(d; t) \end{aligned} \quad (1.2.56)$$

using the expression:

$$\delta_{d, d'} = \frac{1}{2\pi} \int_{-\pi}^{\pi} dk e^{ik(d-d')} \quad (1.2.57)$$

Applying Eq. (1.2.54) to Eq. (1.2.53):

$$\begin{aligned} \tilde{P}(k; t) &= \sum_d e^{-ikd} \sum_{d'} f(d - d') P(d'; t-1) \\ &= \sum_{d'} \sum_d e^{-ik(d-d')} e^{-ikd'} f(d - d') P(d'; t-1) \\ &= \sum_{d'} \sum_{d''} e^{-ikd''} e^{-ikd'} f(d'') P(d'; t-1) \\ &= \sum_{d''} e^{-ikd''} f(d'') \sum_{d'} e^{-ikd'} P(d'; t-1) = \tilde{f}(k) \tilde{P}(k; t-1) \end{aligned} \quad (1.2.58)$$

we can iterate the equation to obtain:

$$\tilde{P}(k;t) = \tilde{f}(k)\tilde{P}(k;t-1) = \tilde{f}(k)^t \quad (1.2.59)$$

where we use the definition $d(1) = s(1)$ that ensures that $P(d;1) = P(s;1) = f(d)$.

For large t the walker has traveled a large distance, so we are interested in variations of the probability $P(d;t)$ over large distances. Thus, in Fourier space we are concerned with small values of k . To simplify Eq. (1.2.59) for large t we expand $\tilde{f}(k)$ near $k=0$. From Eq. (1.2.54) we can directly evaluate the derivatives of $\tilde{f}(k)$ at $k=0$ in terms of averages:

$$\left. \frac{d^n \tilde{f}(k)}{d^n k} \right|_{k=0} = \sum_s (-is)^n f(s) = (-i)^n \langle s^n \rangle \quad (1.2.60)$$

We can use this expression to evaluate the terms of a Taylor expansion of $\tilde{f}(k)$:

$$\tilde{f}(k) = \tilde{f}(0) + \left. \frac{\partial \tilde{f}(k)}{\partial k} \right|_{k=0} k + \frac{1}{2} \left. \frac{\partial^2 \tilde{f}(k)}{\partial k^2} \right|_{k=0} k^2 + \dots \quad (1.2.61)$$

$$\tilde{f}(k) = \langle 1 \rangle - i \langle s \rangle k - \frac{1}{2} \langle s^2 \rangle k^2 + \dots \quad (1.2.62)$$

Using the normalization of the probability ($\langle 1 \rangle = 1$), and Eqs. (1.2.46) and (1.2.47), gives us:

$$\tilde{P}(k;t) = \left(1 - \frac{1}{2} \sigma_0^2 k^2 + \dots \right)^t \quad (1.2.63)$$

We must now remember that a typical value of $d(t)$, from its RMS value, is $\sigma_0 \sqrt{t}$. By the properties of the Fourier transform, this implies that a typical value of k that we must consider in Eq. (1.2.63) varies with time as $1/\sqrt{t}$. The next term in the expansion, cubic in k , would give rise to a term that is smaller by this factor, and therefore becomes unimportant at long times. If we write $k = q/\sqrt{t}$, then it becomes clearer how to write Eq. (1.2.63) using a limiting expression for large t :

$$\tilde{P}(k;t) = \left(1 - \frac{1}{2} \frac{\sigma_0^2 q^2}{t} + \dots \right)^t \sim e^{-\sigma_0^2 q^2 / 2} = e^{-t \sigma_0^2 k^2 / 2} \quad (1.2.64)$$

This Gaussian, when Fourier transformed back to an expression in d , gives us a Gaussian as follows:

$$P(d;t) = \frac{1}{2\pi} \int_{-\pi}^{\pi} dk e^{ikd} e^{-t \sigma_0^2 k^2 / 2} \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} dk e^{ikd} e^{-t \sigma_0^2 k^2 / 2} \quad (1.2.65)$$

We have extended the integral because the decaying exponential becomes narrow as t increases. The integral is performed by completing the square in the exponent, giving:

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} dk e^{-d^2/2t\sigma_0^2} e^{-(t\sigma_0^2 k^2 - 2ikd - d^2/t\sigma_0^2)/2} = \frac{1}{\sqrt{2\pi t\sigma_0^2}} e^{-d^2/2t\sigma_0^2} \quad (1.2.66)$$

or equivalently:

$$P(d;t) = \frac{1}{\sqrt{2\pi\sigma(t)^2}} e^{-d^2/2\sigma(t)^2} \quad (1.2.67)$$

which is the same as Eq. (1.2.39).

Question 1.2.6 Prove the central limit theorem when s takes a continuum of values.

Solution 1.2.6 The proof follows the same course as the integer valued case. We must define the appropriate averages, and the transform. The average of s is still zero, and the mean square displacement is defined similarly:

$$\langle s \rangle = \int ds s f(s) = 0 \quad (1.2.46')$$

$$\langle s^2 \rangle = \int ds s^2 f(s) = \sigma_0^2 \quad (1.2.47')$$

To avoid problems of notation we substitute the variable x for the state variable d :

$$\langle x(t) \rangle = \left\langle \sum_{t'=1}^t s(t') \right\rangle = t \langle s \rangle = 0 \quad (1.2.48')$$

Skipping steps that are the same we find:

$$\langle x(t)^2 \rangle = \left\langle \left(\sum_{t'=1}^t s(t') \right)^2 \right\rangle = \sum_{t'=1}^t \langle s(t')^2 \rangle = t\sigma_0^2 \quad (1.2.51')$$

since $s(t')$ and $s(t'')$ are still independent for $t' \neq t''$. Eq. (1.2.53) is also essentially unchanged:

$$P(x;t) = \int dx' f(x-x') P(x';t-1) \quad (1.2.53')$$

The transform and inverse transform must now be defined using

$$\begin{aligned} \tilde{P}(k;t) &\equiv \int dx e^{-ikx} P(x;t) \\ \tilde{f}(k) &\equiv \int ds e^{-iks} f(s) \end{aligned} \quad (1.2.54')$$

$$P(d;t) = \frac{1}{2\pi} \int dk e^{ikd} \tilde{P}(k;t) \quad (1.2.55')$$

The latter is proved using the properties of the Dirac (continuum) delta function:

$$\begin{aligned} \delta(x-x') &= \frac{1}{2\pi} \int dk e^{ik(x-x')} \\ \int dx' \delta(x-x') g(x') &= g(x) \end{aligned} \quad (1.2.56')$$

where the latter equation holds for an arbitrary function $g(x)$.

The remainder of the derivation carries forward unchanged. ■

1.2.3 Biased random walk

We now return to the simple random walk with binary steps of ± 1 . The model we consider is a random walk that is biased in one direction. Each time a step is taken there is a probability P_+ for a step of $+1$, that is different from the probability P_- for a step of -1 , or:

$$P(s;t) = P_+ \delta_{s,1} + P_- \delta_{s,-1} \quad (1.2.68)$$

$$P(d' | d) = P_+ \delta_{d',d+1} + P_- \delta_{d',d} \quad (1.2.69)$$

where

$$P_+ + P_- = 1 \quad (1.2.70)$$

What is the average distance traveled in time t ?

$$\langle d(t) \rangle = \sum_{t'=1}^t \langle s(t') \rangle = \sum_{t'=1}^t (P_+ - P_-) = t(P_+ - P_-) \quad (1.2.71)$$

This equation justifies defining the mean velocity as

$$v = P_+ - P_- \quad (1.2.72)$$

Since we already have an average displacement it doesn't make sense to also ask for a typical displacement, as we did with the random walk—the typical displacement is the average one. However, we can ask about the spread of the displacements around the average displacement

$$\begin{aligned} \sigma(t)^2 &= \langle (d(t) - \langle d(t) \rangle)^2 \rangle = \langle d(t)^2 \rangle - 2\langle d(t) \rangle^2 + \langle d(t) \rangle^2 \\ &= \langle d(t)^2 \rangle - \langle d(t) \rangle^2 \end{aligned} \quad (1.2.73)$$

This is called the standard deviation and it reduces to the RMS distance in the unbiased case. For many purposes $\sigma(t)$ plays the same role in the biased random walk as in the unbiased random walk. From Eq. (1.2.71) and Eq. (1.2.72) the second term is $(vt)^2$. The first term is:

$$\begin{aligned}
\langle d(t)^2 \rangle &= \left\langle \left(\sum_{t'=1}^t s(t') \right)^2 \right\rangle = \sum_{t',t''=1}^t \langle s(t')s(t'') \rangle \\
&= \sum_{t',t''=1}^t \left(\delta_{t',t''} + (1 - \delta_{t',t''})(P_+^2 + P_-^2 - 2P_+P_-) \right) \\
&= t + t(t-1)v^2 = t^2v^2 + t(1-v^2)
\end{aligned} \tag{1.2.74}$$

Substituting in Eq. (1.2.73):

$$\sigma^2 = t(1-v^2) \tag{1.2.75}$$

It is interesting to consider this expression in the two limits $v = 1$ and $v = 0$. For $v = 1$ the walk is deterministic, $P_+ = 1$ and $P_- = 0$, and there is no element of chance; the walker always walks to the right. This is equivalent to the iterative map Eq. (1.1.4). Our result Eq. (1.2.66) is that $\sigma = 0$, as it must be for a deterministic system. However, for smaller velocities, the spreading of the systems σ increases until at $v = 0$ we recover the case of the unbiased random walk.

The complete probability distribution is given by:

$$P(d;t) = P_+^{(d+t)/2} P_-^{(d-t)/2} \binom{t}{(d+t)/2} \delta_{t,d}^{\text{oddeven}} \tag{1.2.76}$$

For large t the distribution can be found as we did for the unbiased random walk. The work is left to Question 1.2.7.

Question 1.2.7 Find the long time (continuum) distribution for the biased random walk.

Solution 1.2.7 We use the Sterling approximation as before and take the logarithm of the probability. In addition to the expression from the first line of Eq. (1.2.38) we have an additional factor due to the coefficient of Eq. (1.2.76) which appears in place of the factor of $1/2^t$. We again define $x = d/t$, and divide by 2 to allow both odd and even integers. We obtain the expression:

$$\begin{aligned}
\ln(P(d,t)) &= (t/2)[(1+x)\ln 2P_+ + (1-x)\ln 2P_-] \\
&\quad - (t/2)[(1+x)\ln(1+x) + (1-x)\ln(1-x)] - (1/2)\ln(2\pi t(1-x^2))
\end{aligned} \tag{1.2.77}$$

It makes the most sense to expand this around the mean of x , $\langle x \rangle = v$. To simplify the notation we can use Eq. (1.2.70) and Eq. (1.2.72) to write:

$$\begin{aligned}
P_+ &= (1+v)/2 \\
P_- &= (1-v)/2
\end{aligned} \tag{1.2.78}$$

With these substitutions we have:

$$\begin{aligned}
\ln(P(d,t)) &= (t/2)[(1+x)\ln(1+v) + (1-x)\ln(1-v)] \\
&\quad - (t/2)[(1+x)\ln(1+x) + (1-x)\ln(1-x)] - (1/2)\ln(2\pi t(1-x^2))
\end{aligned} \tag{1.2.79}$$

We expand the first two terms in a Taylor expansion around the mean of x and expand the third term inside the logarithm. The first term of Eq. (1.2.79) has only a constant and linear term in a Taylor expansion. These cancel the constant and the first derivative of the Taylor expansion of the second term of Eq. (1.2.79) at $x = v$. Higher derivatives arise only from the second term:

$$\begin{aligned}\ln(P(d,t)) &= -(t/2) \left[\frac{1}{(1-v^2)}(x-v)^2 + \frac{2}{3(1-v^2)^2}(x-v)^3 + K \right] \\ &\quad - (1/2) \ln(2\pi t[(1-v^2) - 2v(x-v) + K]) \\ &= - \left[\frac{(d-vt)^2}{2\sigma(t)^2} + \frac{(d-vt)^3}{3\sigma(t)^4} + K \right] - (1/2) \ln(2\pi(\sigma(t)^2 - 2v(d-vt) + K))\end{aligned}\quad (1.2.80)$$

In the last line we have restored d and used Eq. (1.2.75). Keeping only the first terms in both expansions gives us:

$$P(d;t) = \frac{1}{\sqrt{2\pi\sigma(t)^2}} e^{-(d-vt)^2/2\sigma(t)^2} \quad (1.2.81)$$

which is a Gaussian distribution around the mean we obtained before. This implies that aside from the constant velocity, and a slightly modified standard deviation, the distribution remains unchanged.

The second term in both expansions in Eq. (1.2.80) become small in the limit of large t , as long as we are not interested in the tail of the distribution. Values of $(d - vt)$ relevant to the main part of the distribution are given by the standard deviation, $\sigma(t)$. The second terms in Eq. (1.2.80) are thus reduced by a factor of $\sigma(t)$ compared to the first terms in the series. Since $\sigma(t)$ grows as the square root of the time, they become insignificant for long times. The convergence is slower, however, than in the unbiased random walk (Questions 1.2.2–1.2.5). ■

Question 1.2.8 You are a manager of a casino and are told by the owner that you have a cash flow problem. In order to survive, you have to make sure that nine out of ten working days you have a profit. Assume that the only game in your casino is a roulette wheel. Bets are limited to only red or black with a 2:1 payoff. The roulette wheel has an equal number of red numbers and black numbers and one green number (the house always wins on green). Assume that people make a fixed number of 10^6 total \$1 bets on the roulette wheel in each day.

- What is the maximum number of red numbers on the roulette wheel that will still allow you to achieve your objective?
- With this number of red numbers, how much money do you make on average in each day?

Solution 1.2.8 The casino wins \$1 for every wrong bet and loses \$1 for every right bet. The results of bets at the casino are equivalent to a random walk with a bias given by:

$$P_+ = (N_{red} + 1)/(N_{red} + N_{black} + 1) \quad (1.2.82)$$

$$P_- = N_{black}/(N_{red} + N_{black} + 1) \quad (1.2.83)$$

where, as the manager, we consider positive the wins of the casino. The color subscripts can be used interchangeably, since the number of red and black is equal. The velocity of the random walk is given by:

$$v = 1/(2N_{red} + 1) \quad (1.2.84)$$

To calculate the probability that the casino will lose on a particular day we must sum the probability that the random walk after 10^6 steps will result in a negative number. We approximate the sum by an integral over the distribution of Eq. (1.2.81). To avoid problems of notation we replace d with y :

$$\begin{aligned} P_{loss} &= \int_{-\infty}^0 dy P(y; t = 10^6) = \frac{1}{\sqrt{2\pi\sigma(t)^2}} \int_{-\infty}^0 dy e^{-(y-vt)^2/2\sigma(t)^2} \\ &= \frac{1}{\sqrt{2\pi\sigma(t)^2}} \int_{-\infty}^{-vt} dy' e^{-(y')^2/2\sigma(t)^2} \end{aligned} \quad (1.2.85)$$

$$\begin{aligned} &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{z_0} dz e^{-z^2} = \frac{1}{2} (1 - \text{erf}(z_0)) \\ & \quad z = y'/\sqrt{2\sigma(t)} \\ & \quad z_0 = -vt/\sqrt{2\sigma(t)^2} = -vt/\sqrt{2t(1-v^2)} \end{aligned} \quad (1.2.86)$$

We have written the probability of loss in a day in terms of the error function $\text{erf}(x)$ —the integral of a Gaussian defined by

$$\text{erf}(z_0) \equiv \frac{2}{\sqrt{\pi}} \int_0^{z_0} dz e^{-z^2} \quad (1.2.87)$$

Since

$$\text{erf}(\infty) = 1 \quad (1.2.88)$$

we have the expression

$$(1 - \text{erf}(z_0)) \equiv \frac{2}{\sqrt{\pi}} \int_{z_0}^{\infty} dz e^{-z^2} \quad (1.2.89)$$

which is also known as the complementary error function $\text{erfc}(x)$.

To obtain the desired constraint on the number of red numbers, or equivalently on the velocity, we invert Eq. (1.2.85) to find a value of v that gives the desired $P_{\text{loss}} = 0.1$, or $\text{erf}(z_0) = 0.8$. Looking up the error function or using iterative guessing on an appropriate computer gives $z_0 = 0.9062$. Inverting Eq. (1.2.86) gives:

$$v = \frac{1}{\sqrt{t/2z_0 - 1}} \approx \sqrt{2z_0/t} \quad (1.2.90)$$

The approximation holds because t is large. The numerical result is $v = 0.0013$. This gives us the desired number of each color (inverting Eq. (1.2.84)) of $N_{\text{red}} = 371$. Of course the result is a very large number and the problem of winning nine out of ten days is a very conservative problem for a casino. Even if we insist on winning ninety-nine out of one hundred days we would have $\text{erf}(z_0) = 0.98$, $z_0 = 1.645$, $v = 0.0018$ and $N_{\text{red}} = 275$. The profits per day in each case are given by vt , which is approximately \$1,300 and \$1,800 respectively. Of course this is much less than for bets on a more realistic roulette wheel. Eventually as we reduce the chance of the casino losing and z_0 becomes larger, we might become concerned that we are describing the properties of the tail of the distribution when we calculate the fraction of days the casino might lose, and Eq. (1.2.85) will not be very accurate. However, it is not difficult to see that casinos do not have cash flow problems. ■

In order to generalize the proof of the central limit theorem to the case of a biased random walk, we can treat the continuum case most simply by considering the system variable $\hat{x}$, where (using $d \rightarrow x$ for the continuum case):

$$\hat{x} = x - \langle x \rangle_t = x - t \langle s \rangle = x - vt \quad (1.2.91)$$

Only x is a stochastic variable on the right side, v and t are numbers. Since iterations of this variable would satisfy the conditions for the generalized random walk, the generalization of the Gaussian distribution to Eq. (1.2.81) is proved. The discrete case is more difficult to prove because we cannot shift the variable d by arbitrary amounts and continue to consider it as discrete. We can argue the discrete case to be valid on the basis of the result for the continuum case, but a separate proof can be constructed as well.

1.2.4 Master equation approach

The Master equation is an alternative approach to stochastic systems, an alternative to Eq. (1.2.5), that is usually applied when time is continuous. We develop it starting from the discrete time case. We can rewrite Eq. (1.2.5) in the form of a difference equation for a particular probability $P(s)$. Beginning from:

$$P(s;t) = P(s;t-1) + \left(\sum_{s'} P(s|s')P(s';t-1) - P(s;t-1) \right) \quad (1.2.92)$$

we extract the term where the system remains in the same state:

$$P(s;t) = P(s;t-1) + \left(\sum_{s' \neq s} P(s|s')P(s';t-1) + P(s|s)P(s;t-1) - P(s;t-1) \right) \quad (1.2.93)$$

We use the normalization of probability to write it in terms of the transitions away from this site:

$$P(s;t) = P(s;t-1) + \left(\sum_{s' \neq s} P(s|s')P(s';t-1) + \left(1 - \sum_{s' \neq s} P(s'|s) \right) P(s;t-1) - P(s;t-1) \right) \quad (1.2.94)$$

Canceling the terms in the bracket that refer only to the probability $P(s;t-1)$ we write this as a difference equation. On the right appear only the probabilities at different values of the state variable ($s' \neq s$):

$$P(s;t) - P(s;t-1) = \sum_{s' \neq s} \left(P(s|s')P(s';t-1) - P(s'|s)P(s;t-1) \right) \quad (1.2.95)$$

To write the continuum form we reintroduce the time difference between steps Δt .

$$\frac{P(s;t) - P(s;t-\Delta t)}{\Delta t} = \sum_{s' \neq s} \left(\frac{P(s|s')}{\Delta t} P(s';t-\Delta t) - \frac{P(s'|s)}{\Delta t} P(s;t-\Delta t) \right) \quad (1.2.96)$$

When the limit of $\Delta t \rightarrow 0$ is meaningful, it is possible to make the change to the equation

$$\dot{P}(s;t) = \sum_{s' \neq s} \left(R(s|s')P(s';t) - R(s'|s)P(s;t) \right) \quad (1.2.97)$$

Where the ratio $P(s|s')/\Delta t$ has been replaced by the rate of transition $R(s|s')$. Eq. (1.2.97) is called the Master equation and we can consider Eq. (1.2.95) as the discrete time analog.

The Master equation has a simple interpretation: The rate of change of the probability of a particular state is the total rate at which probability is being added into that state from all other states, minus the total rate at which probability is leaving the state. Probability is acting like a fluid that is flowing to or from a particular state and is being conserved, as it must be. Eq. (1.2.97) is very much like the continuity equation of fluid flow, where the density of the fluid at a particular place changes according to how much is flowing to that location or from it. We will construct and use the Master equation approach to discuss the problem of relaxation in activated processes in Section 1.4.

1.3 Thermodynamics and Statistical Mechanics

The field of thermodynamics is easiest to understand in the context of Newtonian mechanics. Newtonian mechanics describes the effect of forces on objects. Thermodynamics describes the effect of heat transfer on objects. When heat is transferred, the temperature of an object changes. Temperature and heat are also intimately related to energy. A hot gas in a piston has a high pressure and it can do mechanical work by applying a force to a piston. By Newtonian mechanics the work is directly related to a transfer of energy. The laws of Newtonian mechanics are simplest to describe using the abstract concept of a point object with mass but no internal structure. The analogous abstraction for thermodynamic laws are materials that are in equilibrium and (even better) are homogeneous. It turns out that even the description of the equilibrium properties of materials is so rich and varied that this is still a primary focus of active research today.

Statistical mechanics begins as an effort to explain the laws of thermodynamics by considering the microscopic application of Newton's laws. Microscopically, the temperature of a gas is found to be related to the kinetic motion of the gas molecules. Heat transfer is the transfer of Newtonian energy from one object to another. The statistical treatment of the many particles of a material, with a key set of assumptions, reveals that thermodynamic laws are a natural consequence of many microscopic particles interacting with each other. Our studies of complex systems will lead us to discuss the properties of systems composed of many interacting parts. The concepts and tools of statistical mechanics will play an important role in these studies, as will the laws of thermodynamics that emerge from them. Thermodynamics also begins to teach us how to think about systems interacting with each other.

1.3.1 Thermodynamics

Thermodynamics describes macroscopic pieces of material in equilibrium in terms of macroscopic parameters. Thermodynamics was developed as a result of experience/experiment and, like Newton's laws, is to be understood as a set of self-consistent definitions and equations. As with Newtonian mechanics, where in its simplest form objects are point particles and friction is ignored, the discussion assumes an idealization that is directly experienced only in special circumstances. However, the fundamental laws, once understood, can be widely applied. The central quantities that are to be defined and related are the energy U , temperature T , entropy S , pressure P , the mass (which we write as the number of particles) N , and volume V . For magnets, the quantities should include the magnetization M , and the magnetic field H . Other macroscopic quantities that are relevant may be added as necessary within the framework developed by thermodynamics. Like Newtonian mechanics, a key aspect of thermodynamics is to understand how systems can be acted upon or can act upon each other. In addition to the quantities that describe the state of a system, there are two quantities that describe actions that may be made on a system to change its state: work and heat transfer.

The equations that relate the macroscopic quantities are known as the zeroth, first and second laws of thermodynamics. Much of the difficulty in understanding thermodynamics arises from the way the entropy appears as an essential but counter-intuitive quantity. It is more easily understood in the context of a statistical treatment included below. A second source of difficulty is that even a seemingly simple material system, such as a piece of metal in a room, is actually quite complicated thermodynamically. Under usual circumstances the metal is not in equilibrium but is emitting a vapor of its own atoms. A thermodynamic treatment of the metal requires consideration not only of the metal but also the vapor and even the air that applies a pressure upon the metal. It is therefore generally simplest to consider the thermodynamics of a gas confined in a closed (and inert) chamber as a model thermodynamic system. We will discuss this example in detail in Question 1.3.1. The translational motion of the whole system, treated by Newtonian mechanics, is ignored.

We begin by defining the concept of equilibrium. A system left in isolation for a long enough time achieves a macroscopic state that does not vary in time. The system in an unchanging state is said to be in equilibrium. Thermodynamics also relies upon a particular type of equilibrium known as thermal equilibrium. Two systems can be brought together in such a way that they interact only by transferring heat from one to the other. The systems are said to be in thermal contact. An example would be two gases separated by a fixed but thermally conducting wall. After a long enough time the system composed of the combination of the two original systems will be in equilibrium. We say that the two systems are in thermal equilibrium with each other. We can generalize the definition of thermal equilibrium to include systems that are not in contact. We say that any two systems are in thermal equilibrium with each other if they do not change their (macroscopic) state when they are brought into thermal contact. Thermal equilibrium does not imply that the system is homogeneous, for example, the two gases may be at different pressures.

The zeroth law of thermodynamics states that if two systems are in thermal equilibrium with a third they are in thermal equilibrium with each other. This is not obvious without experience with macroscopic objects. The zeroth law implies that the interaction that occurs during thermal contact is not specific to the materials, it is in some sense weak, and it matters not how many or how big are the systems that are in contact. It enables us to define the temperature T as a quantity which is the same for all systems in thermal equilibrium. A more specific definition of the temperature must wait till the second law of thermodynamics. We also define the concept of a thermal reservoir as a very large system such that any system that we are interested in, when brought into contact with the thermal reservoir, will change its state by transferring heat to or from the reservoir until it is in equilibrium with the reservoir, but the transfer of heat will not affect the temperature of the reservoir.

Quite basic to the formulation and assumptions of thermodynamics is that the macroscopic state of an isolated system in equilibrium is completely defined by a specification of three parameters: energy, mass and volume (U, N, V). For magnets we must add the magnetization M ; we will leave this case for later. The confinement of

the system to a volume V is understood to result from some form of containment. The state of a system can be characterized by the force per unit area—the pressure P —exerted by the system on the container or by the container on the system, which are the same. Since in equilibrium a system is uniquely described by the three quantities (U, N, V) , these determine all the other quantities, such as the pressure P and temperature T . Strictly speaking, temperature and pressure are only defined for a system in equilibrium, while the quantities (U, N, V) have meaning both in and out of equilibrium.

It is assumed that for a homogeneous material, changing the size of the system by adding more material in equilibrium at the same pressure and temperature changes the mass, number of particles N , volume V and energy U , in direct proportion to each other. Equivalently, it is assumed that cutting the system into smaller parts results in each subpart retaining the same properties in proportion to each other (see Figs. 1.3.1 and 1.3.2). This means that these quantities are additive for different parts of a system whether isolated or in thermal contact or full equilibrium:

$$\begin{aligned} N &= \sum_{\alpha} N^{\alpha} \\ V &= \sum_{\alpha} V^{\alpha} \\ U &= \sum_{\alpha} U^{\alpha} \end{aligned} \tag{1.3.1}$$

where α indexes the parts of the system. This would not be true if the parts of the system were strongly interacting in such a way that the energy depended on the relative location of the parts. Properties such as (U, N, V) that are proportional to the size of the system are called extensive quantities. Intensive quantities are properties that do not change with the size of the system at a given pressure and temperature. The ratio of two extensive quantities is an intensive quantity. Examples are the particle density N/V and the energy density U/V . The assumption of the existence of extensive and intensive quantities is also far from trivial, and corresponds to the intuition that for a macroscopic object, the local properties of the system do not depend on the size of the system. Thus a material may be cut into two parts, or a small part may be separated from a large part, without affecting its local properties.

The simplest thermodynamic systems are homogeneous ones, like a gas in an inert container. However we can also use Eq. (1.3.1) for an inhomogeneous system. For example, a sealed container with water inside will reach a state where both water and vapor are in equilibrium with each other. The use of intensive quantities and the proportionality of extensive quantities to each other applies only within a single phase—a single homogeneous part of the system, either water or vapor. However, the additivity of extensive quantities in Eq. (1.3.1) still applies to the whole system. A homogeneous as well as a heterogeneous system may contain different chemical species. In this case the quantity N is replaced by the number of each chemical species N_i ; and the first line of Eq. (1.3.1) may be replaced by a similar equation for each species.

Figure 1.3.1 Thermodynamics considers macroscopic materials. A basic assumption is that cutting a system into two parts will not affect the local properties of the material and that the energy U , mass (or number of particles) N and the volume V will be divided in the same proportion. The process of separation is assumed to leave the materials under the same conditions of pressure and temperature. ■

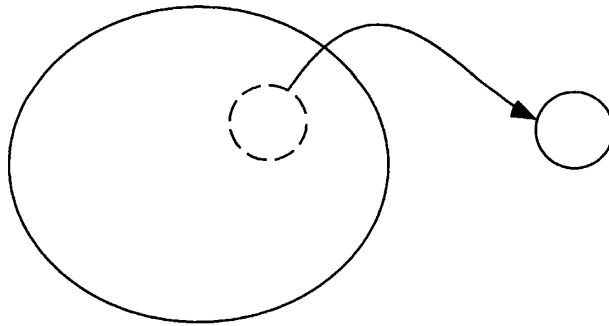
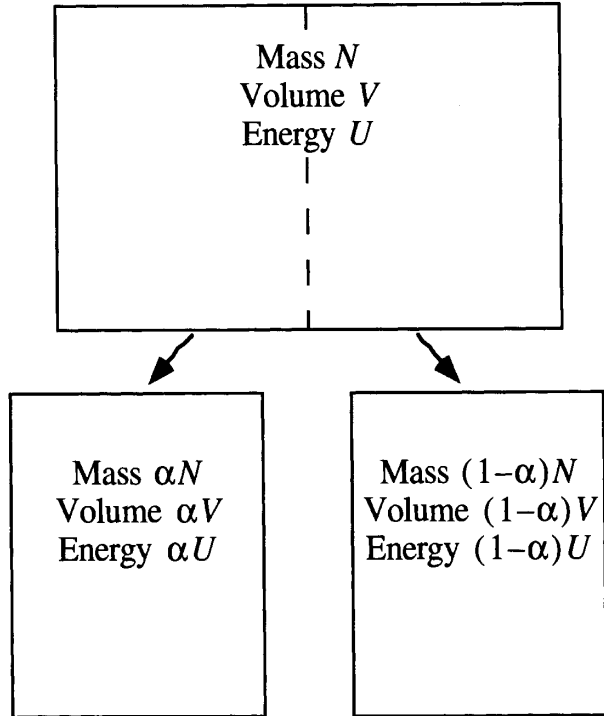


Figure 1.3.2 The assumption that the local properties of a system are unaffected by subdivision applies also to the case where a small part of a much larger system is removed. The local properties, both of the small system and of the large system are assumed to remain unchanged. Even though the small system is much smaller than the original system, the small system is understood to be a macroscopic piece of material. Thus it retains the same local properties it had as part of the larger system. ■

The first law of thermodynamics describes how the energy of a system may change. The energy of an isolated system is conserved. There are two macroscopic processes that can change the energy of a system when the number of particles is fixed.

The first is work, in the sense of applying a force over a distance, such as driving a piston that compresses a gas. The second is heat transfer. This may be written as:

$$dU = q + w \quad (1.3.2)$$

where q is the heat transfer into the system, w is the work done on the system and U is the internal energy of the system. The differential d signifies the incremental change in the quantity U as a result of the incremental process of heat transfer and work. The work performed on a gas (or other system) is the force times the distance applied Fdx , where we write F as the magnitude of the force and dx as an incremental distance. Since the force is the pressure times the area $F = PA$, the work is equal to the pressure times the volume change or:

$$w = -PAdx = -PdV \quad (1.3.3)$$

The negative sign arises because positive work on the system, increasing the system's energy, occurs when the volume change is negative. Pressure is defined to be positive.

If two systems act upon each other, then the energy transferred consists of both the work and heat transfer. Each of these are separately equal in magnitude and opposite in sign:

$$\begin{aligned} dU_1 &= q_{21} + w_{21} \\ dU_2 &= q_{12} + w_{12} \\ q_{12} &= -q_{21} \\ w_{12} &= -w_{21} \end{aligned} \quad (1.3.4)$$

where q_{21} is the heat transfer from system 2 to system 1, and w_{21} is the work performed by system 2 on system 1. q_{12} and w_{12} are similarly defined. The last line of Eq. (1.3.4) follows from Newton's third law. The other equations follow from setting $dU = 0$ (Eq. (1.3.2)) for the total system, composed of both of the systems acting upon each other.

The second law of thermodynamics given in the following few paragraphs describes a few key aspects of the relationship of the equilibrium state with nonequilibrium states. The statement of the second law is essentially a definition and description of properties of the entropy. Entropy enables us to describe the process of approach to equilibrium. In the natural course of events, any system in isolation will change its state toward equilibrium. A system which is not in equilibrium must therefore undergo an irreversible process leading to equilibrium. The process is irreversible because the reverse process would take us away from equilibrium, which is impossible for a macroscopic system. Reversible change can occur if the state of a system in equilibrium is changed by transfer of heat or by work in such a way (slowly) that it always remains in equilibrium.

For every macroscopic state of a system (not necessarily in equilibrium) there exists a quantity S called the entropy of the system. The change in S is positive for any natural process (change toward equilibrium) of an isolated system

$$dS \geq 0 \quad (1.3.5)$$

For an isolated system, equality holds only in equilibrium when no change occurs. The converse is also true—any possible change that increases S is a natural process. Therefore, for an isolated system S achieves its maximum value for the equilibrium state.

The second property of the entropy describes how it is affected by the processes of work and heat transfer during reversible processes. The entropy is affected only by heat transfer and not by work. If we only perform work and do not transfer heat the entropy is constant. Such processes where $q = 0$ are called adiabatic processes. For adiabatic processes $dS = 0$.

The third property of the entropy is that it is extensive:

$$S = \sum_{\alpha} S^{\alpha} \quad (1.3.6)$$

Since in equilibrium the state of the system is defined by the macroscopic quantities (U, N, V) , S is a function of them— $S = S(U, N, V)$ —in equilibrium. The fourth property of the entropy is that if we keep the size of the system constant by fixing both the number of particles N and the volume V , then the change in entropy S with increasing energy U is always positive:

$$\left(\frac{\partial S}{\partial U} \right)_{N,V} > 0 \quad (1.3.7)$$

where the subscripts denote the (values of the) constant quantities. Because of this we can also invert the function $S = S(U, N, V)$ to obtain the energy U in terms of S , N and V : $U = U(S, N, V)$.

Finally, we mention that the zero of the entropy is arbitrary in classical treatments. The zero of entropy does attain significance in statistical treatments that include quantum effects.

Having described the properties of the entropy for a single system, we can now reconsider the problem of two interacting systems. Since the entropy describes the process of equilibration, we consider the process by which two systems equilibrate thermally. According to the zeroth law, when the two systems are in equilibrium they are at the same temperature. The two systems are assumed to be isolated from any other influence, so that together they form an isolated system with energy U_t and entropy S_t . Each of the subsystems is itself in equilibrium, but they are at different temperatures initially, and therefore heat is transferred to achieve equilibrium. The heat transfer is assumed to be performed in a reversible fashion—slowly. The two subsystems are also assumed to have a fixed number of particles N_1, N_2 and volume V_1, V_2 . No work is done, only heat is transferred. The energies of the two systems U_1 and U_2 and entropies S_1 and S_2 are not fixed.

The transfer of heat results in a transfer of energy between the two systems according to Eq. (1.3.4), since the total energy

$$U_t = U_1 + U_2 \quad (1.3.8)$$

is conserved, we have

$$dU_t = dU_1 + dU_2 = 0 \quad (1.3.9)$$

We will consider the processes of equilibration twice. The first time we will identify the equilibrium condition and the second time we will describe the equilibration. At equilibrium the entropy of the whole system is maximized. Variation of the entropy with respect to any internal parameter will give zero at equilibrium. We can consider the change in the entropy of the system as a function of how much of the energy is allocated to the first system:

$$\frac{dS_t}{dU_1} = \frac{dS_1}{dU_1} + \frac{dS_2}{dU_1} = 0 \quad (1.3.10)$$

in equilibrium. Since the total energy is fixed, using Eq. (1.3.9) we have:

$$\frac{dS_t}{dU_1} = \frac{dS_1}{dU_1} - \frac{dS_2}{dU_2} = 0 \quad (1.3.11)$$

or

$$\frac{dS_1}{dU_1} = \frac{dS_2}{dU_2} \quad (1.3.12)$$

in equilibrium. By the definition of the temperature, any function of the derivative of the entropy with respect to energy could be used as the temperature. It is conventional to define the temperature T using:

$$\frac{1}{T} = \left(\frac{dS}{dU} \right)_{N,V} \quad (1.3.13)$$

This definition corresponds to the Kelvin temperature scale. The units of temperature also define the units of the entropy. This definition has the advantage that heat always flows from the system at higher temperature to the system at lower temperature.

To prove this last statement, consider a natural small transfer of heat from one system to the other. The transfer must result in the two systems raising their collective entropy:

$$dS_t = dS_1 + dS_2 \geq 0 \quad (1.3.14)$$

We rewrite the change in entropy of each system in terms of the change in energy. We recall that N and V are fixed for each of the two systems and the entropy is a function only of the three macroscopic parameters (U, N, V) . The change in S for each system may be written as:

$$\begin{aligned} dS_1 &= \left(\frac{\partial S}{\partial U} \right)_{N_1, V_1} dU_1 \\ dS_2 &= \left(\frac{\partial S}{\partial U} \right)_{N_2, V_2} dU_2 \end{aligned} \quad (1.3.15)$$

to arrive at:

$$\left(\frac{\partial S}{\partial U}\right)_{N_1, V_1} dU_1 + \left(\frac{\partial S}{\partial U}\right)_{N_2, V_2} dU_2 \geq 0 \quad (1.3.16)$$

or using Eq. (1.3.9) and the definition of the temperature (Eq. (1.3.13)) we have:

$$\left[\left(\frac{1}{T_1}\right) - \left(\frac{1}{T_2}\right)\right] dU_1 \geq 0 \quad (1.3.17)$$

or:

$$(T_2 - T_1) dU_1 \geq 0 \quad (1.3.18)$$

This implies that a natural process of heat transfer results in the energy of the first system increasing ($dU_1 > 0$) if the temperature of the second system is greater than the first ($(T_2 - T_1) > 0$), or conversely, if the temperature of the second system is less than the temperature of the first.

Using the definition of temperature, we can also rewrite the expression for the change in the energy of a system due to heat transfer or work, Eq. (1.3.2). The new expression is restricted to reversible processes. As in Eq. (1.3.2), N is still fixed. Considering only reversible processes means we consider only equilibrium states of the system, so we can write the energy as a function of the entropy $U = U(S, N, V)$. Since a reversible process changes the entropy and volume while keeping this function valid, we can write the change in energy for a reversible process as

$$\begin{aligned} dU &= \left(\frac{\partial U}{\partial S}\right)_{N, V} dS + \left(\frac{\partial U}{\partial V}\right)_{N, S} dV \\ &= TdS + \left(\frac{\partial U}{\partial V}\right)_{N, S} dV \end{aligned} \quad (1.3.19)$$

The first term reflects the effect of a change in entropy and the second reflects the change in volume. The change in entropy is related to heat transfer but not to work. If work is done and no heat is transferred, then the first term is zero. Comparing the second term to Eq. (1.3.2) we find

$$P = -\left(\frac{\partial U}{\partial V}\right)_{N, S} \quad (1.3.20)$$

and the incremental change in energy for a reversible process can be written:

$$dU = TdS - PdV \quad (1.3.21)$$

This relationship enables us to make direct experimental measurements of entropy changes. The work done on a system, in a reversible or irreversible process, changes the energy of the system by a known amount. This energy can then be extracted in a reversible process in the form of heat. When the system returns to its original state, we

can quantify the amount of heat transferred as a form of energy. Measured heat transfer can then be related to entropy changes using $q = TdS$.

Our treatment of the fundamentals of thermodynamics was brief and does not contain the many applications necessary for a detailed understanding. The properties of S that we have described are sufficient to provide a systematic treatment of the thermodynamics of macroscopic bodies. However, the entropy is more understandable from a microscopic (statistical) description of matter. In the next section we introduce the statistical treatment that enables contact between a microscopic picture and the macroscopic thermodynamic treatment of matter. We will use it to give microscopic meaning to the entropy and temperature. Once we have developed the microscopic picture we will discuss two applications. The first application, the ideal gas, is discussed in Section 1.3.3. The discussion of the second application, the Ising model of magnetic systems, is postponed to Section 1.6.

1.3.2 *The macroscopic state from microscopic statistics*

In order to develop a microscopic understanding of the macroscopic properties of matter we must begin by restating the nature of the systems that thermodynamics describes. Even when developing a microscopic picture, the thermodynamic assumptions are relied upon as guides. Macroscopic systems are assumed to have an extremely large number N of individual particles (e.g., at a scale of 10^{23}) in a volume V . Because the size of these systems is so large, they are typically investigated by considering the limit of $N \rightarrow \infty$ and $V \rightarrow \infty$, while the density $n = N/V$ remains constant. This is called the thermodynamic limit. Various properties of the system are separated into extensive and intensive quantities. Extensive quantities are proportional to the size of the system. Intensive quantities are independent of the size of the system. This reflects the intuition that local properties of a macroscopic object do not depend on the size of the system. As in Figs. 1.3.1 and 1.3.2, the system may be cut into two parts, or a small part may be separated from a large part without affecting its local properties.

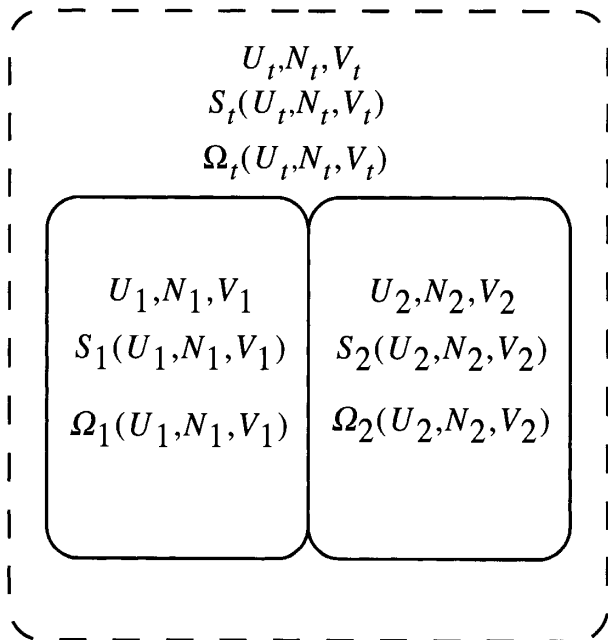
The total energy U of an isolated system in equilibrium, along with the number of particles N and volume V , defines the macroscopic state (macrostate) of an isolated system in equilibrium. Microscopically, the energy of the system E is given in classical mechanics in terms of the complete specification of the individual particle positions, momenta and interaction potentials. Together these define the microscopic state (microstate) of the system. The microstate is defined differently in quantum mechanics but similar considerations apply. When we describe the system microscopically we use the notation E rather than U to describe the energy. The reason for this difference is that macroscopically the energy U has some degree of fuzziness in its definition, though the degree of fuzziness will not enter into our considerations. Moreover, U may also be used to describe the energy of a system that is in thermal equilibrium with another system. However, thinking microscopically, the energy of such a system is not well defined, since thermal contact allows the exchange of energy between the two systems. We should also distinguish between the microscopic and macroscopic concepts of the number of particles and the volume, but since we will not make use of this distinction, we will not do so.

There are many possible microstates that correspond to a particular macrostate of the system specified only by U, N, V . We now make a key assumption of statistical mechanics—that all of the possible microstates of the system occur with equal probability. The number of these microstates $\Omega(U, N, V)$, which by definition depends on the macroscopic parameters, turns out to be central to statistical mechanics and is directly related to the entropy. Thus it determines many of the thermodynamic properties of the system, and can be discussed even though we are not always able to obtain it explicitly.

We consider again the problem of interacting systems. As before, we consider two systems (Fig. 1.3.3) that are in equilibrium separately, with state variables (U_1, N_1, V_1) and (U_2, N_2, V_2) . The systems have a number of microstates $\Omega_1(U_1, N_1, V_1)$ and $\Omega_2(U_2, N_2, V_2)$ respectively. It is not necessary that the two systems be formed of the same material or have the same functional form of $\Omega(U, N, V)$, so the function Ω is also labeled by the system index. The two systems interact in a limited way, so that they can exchange only energy. The number of particles and volume of each system remains fixed. Conservation of energy requires that the total energy $U_t = U_1 + U_2$ remains fixed, but energy may be transferred from one system to the other. As before, our objective is to identify when energy transfer stops and equilibrium is reached.

Consider the number of microstates of the whole system Ω_t . This number is a function not only of the total energy of the system but also of how the energy is allocated between the systems. So, we write $\Omega_t(U_1, U_2)$, and we assume that at any time the energy of each of the two systems is well defined. Moreover, the interaction between the two systems is sufficiently weak so that the number of states of each system

Figure 1.3.3 Illustration of a system formed out of two parts. The text discusses this system when energy is transferred from one part to the other. The transfer of energy on a microscopic scale is equivalent to the transfer of heat on a macroscopic scale, since the two systems are not allowed to change their number of particles or their volume. ■



may be counted independently. Then the total number of microstates is the product of the number of microstates of each of the two systems separately.

$$\Omega_t(U_1, U_2) = \Omega_1(U_1)\Omega_2(U_2) \quad (1.3.22)$$

where we have dropped the arguments N and V , since they are fixed throughout this discussion. When energy is transferred, the number of microstates of each of the two systems is changed. When will the transfer of energy stop? Left on its own, the system will evolve until it reaches the most probable separation of energy. Since any particular state is equally likely, the most probable separation of energy is the separation that gives rise to the greatest possible number of states. When the number of particles is large, the greatest number of states corresponding to a particular energy separation is much larger than the number of states corresponding to any other possible separation. Thus any other possibility is completely negligible. No matter when we look at the system, it will be in a state with the most likely separation of the energy. For a macroscopic system, it is impossible for a spontaneous transfer of energy to occur that moves the system away from equilibrium.

The last paragraph implies that the transfer of energy from one system to the other stops when Ω_t reaches its maximum value. Since $U_t = U_1 + U_2$ we can find the maximum value of the number of microstates using:

$$\begin{aligned} \frac{\partial \Omega_t(U_1, U_t - U_1)}{\partial U_1} = 0 &= \frac{\partial \Omega_1(U_1)}{\partial U_1} \Omega_2(U_t - U_1) + \Omega_1(U_1) \frac{\partial \Omega_2(U_t - U_1)}{\partial U_1} \\ 0 &= \frac{\partial \Omega_1(U_1)}{\partial U_1} \Omega_2(U_2) - \Omega_1(U_1) \frac{\partial \Omega_2(U_2)}{\partial U_2} \end{aligned} \quad (1.3.23)$$

or

$$\begin{aligned} \frac{1}{\Omega_1(U_1)} \frac{\partial \Omega_1(U_1)}{\partial U_1} &= \frac{1}{\Omega_2(U_2)} \frac{\partial \Omega_2(U_2)}{\partial U_2} \\ \frac{\partial \ln \Omega_1(U_1)}{\partial U_1} &= \frac{\partial \ln \Omega_2(U_2)}{\partial U_2} \end{aligned} \quad (1.3.24)$$

The equivalence of these quantities is analogous to the equivalence of the temperature of the two systems in equilibrium. Since the derivatives in the last equation are performed at constant N and V , it appears, by analogy to Eq. (1.3.12), that we can identify the entropy as:

$$S = k \ln(\Omega(E, N, V)). \quad (1.3.25)$$

The constant k , known as the Boltzmann constant, is needed to ensure correspondence of the microscopic counting of states with the macroscopic units of the entropy, as defined by the relationship of Eq. (1.3.13), once the units of temperature and energy are defined.

The entropy as defined by Eq. (1.3.25) can be shown to satisfy all of the properties of the thermodynamic entropy in the last section. We have argued that an isolated

system evolves its macrostate in such a way that it maximizes the number of microstates that correspond to the macrostate. By Eq. (1.3.25), this is the same as the first property of the entropy in Eq. (1.3.5), the maximization of the entropy in equilibrium.

Interestingly, demonstrating the second property of the entropy, that it does not change during an adiabatic process, requires further formal developments relating entropy to information that will be discussed in Sections 1.7 and 1.8. We will connect the two discussions and thus be able to demonstrate the second property of the entropy in Chapter 8 (Section 8.3.2).

The extensive property of the entropy follows from Eq. (1.3.22). This also means that the number of states at a particular energy grows exponentially with the size of the system. More properly, we can say that experimental observation that the entropy is extensive suggests that the interaction between macroscopic materials, or parts of a single macroscopic material, is such that the microstates of each part of the system may be enumerated independently.

The number of microstates can be shown by simple examples to increase with the energy of the system. This corresponds to Eq. (1.3.7). There are also examples where this can be violated, though this will not enter into our discussions.

We consider next a second example of interacting systems that enables us to evaluate the meaning of a system in equilibrium with a reservoir at a temperature T . We consider a small part of a much larger system (Fig. 1.3.4). No assumption is necessary regarding the size of the small system; it may be either microscopic or macroscopic. Because of the contact of the small system with the large system, its energy is not

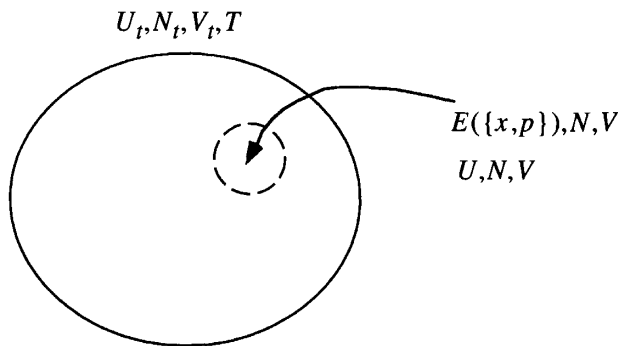


Figure 1.3.4 In order to understand temperature we consider a closed system composed of a large and small system, or equivalently a small system which is part of a much larger system. The larger system serves as a thermal reservoir transferring energy to and from the small system without affecting its own temperature. A microscopic description of this process in terms of a single microscopic state of the small system leads to the Boltzmann probability. An analysis in terms of the macroscopic state of the small system leads to the principle of minimization of the free energy to obtain the equilibrium state of a system at a fixed temperature. This principle replaces the principle of maximization of the entropy, which only applies for a closed system. ■

always the same. Energy will be transferred back and forth between the small and large systems. The essential assumption is that the contact between the large and small system does not affect any other aspect of the description of the small system. This means that the small system is in some sense independent of the large system, despite the energy transfer. This is true if the small system is itself macroscopic, but it may also be valid for certain microscopic systems. We also assume that the small system and the large system have fixed numbers of particles and volumes.

Our objective is to consider the probability that a particular microstate of the small system will be realized. A microstate is identified by all of the microscopic parameters necessary to completely define this state. We use the notation $\{x, p\}$ to denote these coordinates. The probability that this particular state will be realized is given by the fraction of states of the whole system for which the small system attains this state. Because there is only one such state for the small system, the probability that this state will be realized is given by (proportional to) a count of the number of states of the rest of the system. Since the large system is macroscopic, we can count this number by using the macroscopic expression for the number of states of the large system:

$$P(\{x, p\}) \propto \Omega_R(U_t - E(\{x, p\}), N_t - N, V_t - V) \quad (1.3.26)$$

where $E(\{x, p\}), N, V$ are the energy, number of particles and volume of the microscopic system respectively. $E(\{x, p\})$ is a function of the microscopic parameters $\{x, p\}$. U_t, N_t, V_t are the energy, number of particles and volume of the whole system, including both the small and large systems. Ω_R is the entropy of the large subsystem (reservoir). Since the number of states generally grows faster than linearly as a function of the energy, we use a Taylor expansion of its logarithm (or equivalently a Taylor expansion of the entropy) to find

$$\begin{aligned} & \ln \Omega_R(U_t - E(\{x, p\}), N_t - N, V_t - V) \\ &= \ln \Omega_R(U_t, N_t - N, V_t - V) + \left(\frac{\partial \ln \Omega_R(U_t, N_t - N, V_t - V)}{\partial E_t} \right)_{N_t, V_t} (-E(\{x, p\})) \\ &= \ln \Omega_R(U_t, N_t - N, V_t - V) + \frac{1}{kT} (-E(\{x, p\})) \end{aligned} \quad (1.3.27)$$

where we have not expanded in the number of particles and the volume because they are unchanging. We take only the first term in the expansion, because the size of the small system is assumed to be much smaller than the size of the whole system. Exponentiating gives the relative probability of this particular microscopic state.

$$\Omega_R(U_t - E(\{x, p\}), N_t - N, V_t - V) = \Omega_R(U_t, N_t - N, V_t - V) e^{-E(\{x, p\})/kT} \quad (1.3.28)$$

The probability of this particular state must be normalized so that the sum over all states is one. Since we are normalizing the probability anyway, the constant coefficient does not affect the result. This gives us the Boltzmann probability distribution: